

Visual Agentic Memory: Enabling Online Long Video Understanding via Online Indexing, Hierarchical Memory, and Agentic Retrieval

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Abstract

Online long video understanding requires more than long context windows. It also requires decisions, as the stream unfolds, about which visual evidence to retain, how to preserve its searchability, and how to ground later reasoning in recoverable observations rather than compressed latent state alone. Existing systems still struggle to balance these requirements. Bounded-context streaming methods, compact memory schemes, and query-time agents each address part of the problem, but none fully combine efficient online processing with long-horizon evidence recoverability. This dissertation proposes Visual Agentic Memory (VAM), a training-free framework built on online indexing, hierarchical memory, and agentic retrieval.

VAM comprises three components: *Online Indexing*, *Hierarchical Memory*, and *Agentic Retrieval*. Online Indexing selectively retains visual evidence under streaming constraints. Hierarchical Memory stores this evidence in a *Parallel Representation* that aligns temporal context with spatial observations, including de-duplicated and processed raw frames for later visual inspection. Agentic Retrieval then uses a Multimodal Large Language Model (MLLM) to search, inspect, and verify candidate evidence before producing a grounded answer.

On OVO-Bench, which evaluates timestamp-conditioned understanding under streaming constraints, VAM achieves 61.95% overall, the strongest result among the compared open-source agentic systems and frameworks. On MM-Lifelong *train@month*, which evaluates retrieval and reasoning over extended real-world horizons, VAM reaches 17.11% answer accuracy, remaining close to the strongest reported result. These findings support the central claim of this dissertation: online long video understanding benefits from treating visual memory as an explicit, inspectable, and queryable substrate rather than compressed latent state alone.

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1 Introduction

1.1 Motivation and Problem Statement

A fundamental contradiction defines contemporary visual perception. High-bandwidth video acquisition is now a practical reality, yet transforming these unbounded streams into a stable, queryable, and verifiable memory substrate remains an open challenge. This tension between passive recording and active intelligence defines the central difficulty of online long video understanding. While contemporary models can capture vast amounts of visual experience, they struggle to maintain a usable account of what happened, where it happened, and how scenes evolve over time [1, 2, 3, 4].

This requirement is particularly critical for the development of world models [5, 6, 7, 8, 9], which depend on coherent long-range prediction and interaction [10, 11, 12, 13]. A model cannot reason about the future or explain the past if earlier observations cannot be retained, localised, and revisited in a reliable form. In online streaming settings, this implies that understanding depends not only on learning compact state abstractions [14, 15, 16, 17], but also on preserving recoverable evidence across extended durations. The practical challenge, therefore, is how archives ranging from hours to months can remain searchable and auditable under real-world constraints. Beyond task-level accuracy, a robust system must bridge the gap between raw pixels and actionable recall, especially when critical evidence is buried within hours of semantically dense but visually redundant footage [18, 19, 20, 21, 22].

This gap exposes the fundamental limitations of contemporary multimodal architectures: many existing models struggle to maintain useful context over online long video streams and to identify relevant moments reliably at retrieval time [20, 23, 24, 25, 26, 27]. To address this, we frame online long video understanding as an integrated orchestration of indexing, storage, and retrieval, involving: (i) online indexing across temporal and spatial dimensions [28, 29], (ii) hierarchical memory that keeps timeline context and raw visual evidence available together [20, 30], and (iii) task-relevant reasoning through agentic re-

trieval [31, 32, 33, 34]. This perspective also distinguishes retrieval-first agentic systems from recursive agentic approaches such as ReMA [3], which place greater emphasis on iterative reasoning than on explicit evidence retrieval.

Online long-horizon settings introduce fundamental challenges that existing technical paradigms fail to address:

Online and Real-time Processing Challenges:

- **Streaming Visual Redundancy and Memory Saturation:** High-bandwidth video streams are dominated by visually redundant intervals that lead to rapid memory saturation. While offline methods benefit from global temporal smoothing, online systems must perform *Online Indexing* under strict streaming constraints. Fixed-rate sampling methods frequently fail to capture transient visual events or result in the irrecoverable loss of critical state transitions [28, 29, 31].
- **Computational Coupling in Always-on Sensing:** Continuous multimodal sensing requires background online indexing to run concurrently with reasoning. Traditional tightly coupled architectures suffer from processing spikes, where intensive retrieval operations block the indexing of incoming frames, leading to a loss of temporal continuity in always-on applications [34, 35, 36, 37].

Online Long-horizon and Large-scale Retrieval Challenges:

- **Global Localisation Collapse and Ambiguity:** In archives spanning days or months, retrieval precision (signal-to-noise ratio) degrades as the memory pool expands. Standard nearest-neighbour retrieval over global embeddings often fails to disambiguate between repeated daily routines and visually similar scenes, a phenomenon known as global localisation collapse, necessitating more structured hierarchical memory and time-aware retrieval [3, 38, 39].
- **Attention Dilution and Memory Drift:** End-to-end models with large context windows suffer from *Attention Dilution* as duration increases, where noise accumulation prevents the model from focusing on task-relevant moments [25, 26]. Conversely, recursive agentic methods that do not preserve explicit retrievable evidence are susceptible to irreversible *Memory Drift*, where incorrect inferences propagate without the possibility of retrospective inspection against raw evidence [3].

These challenges become more severe when supervision is limited and models must generalise without task-specific retraining. In such cases, strong pre-trained multimodal models provide broad semantic priors, but they do not automatically solve online long-horizon indexing, temporal disambiguation, or provenance tracking [40, 41, 42, 43]. The central requirement is therefore not merely stronger perception, but an online architecture that can update memory continuously under streaming input while preserving temporal grounding, retrieval quality, and access to recoverable evidence over long horizons. In the framework developed here, a *Multimodal Large Language Model* (MLLM) serves as the common multimodal reasoning model for both event summarisation during online indexing and fine-grained visual inspection at query time. In the current implementation, event summarisation is driven by the event’s temporally bounded visual evidence: the MLLM receives the event video clip when available, together with representative retained frames and temporal metadata, rather than relying on text-only abstraction.

To address these constraints, we propose the “Visual Agentic Memory” (VAM) framework, an online, training-free system that transforms raw video streams into hierarchical memory for long-horizon reasoning. Rather than relying only on a flat non-parametric store, VAM organises retained moments and events through a parallel representation across temporal and spatial views. Drawing upon hierarchical memory, online indexing, and retrieval-based agentic reasoning principles [20, 28, 31], the framework supports continuous online indexing and reasoning over unbounded temporal scales. The memory is updated incrementally as new frames arrive, avoiding expensive global preprocessing such as optical flow [28, 35]. As a training-free system, it leverages pre-trained vision-language models for zero-shot visual understanding [40, 41, 42]. At the representation level, VAM groups moments into events, where each event maintains a temporal representation for timeline reasoning and a spatial representation for frame-level evidence, including de-duplicated and processed raw frames preserved for later visual inspection. Event documents are written from the same temporally bounded evidence: when available, the MLLM receives the event video clip together with representative retained frames and temporal metadata. Together, these form a parallel representation that supports both context-aware search and raw visual lookup. For complex queries, the retrieval agent performs multi-turn search and visual inspection over both representations until it can answer with precisely cited intervals [32, 44].

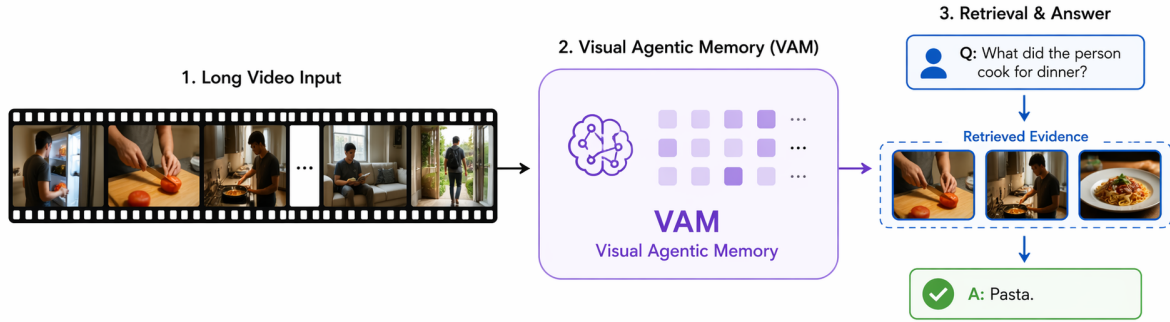


Figure 1.1: Visual Agentic Memory (VAM) turns continuous video streams into searchable long-horizon memory. Unlike prior offline methods that operate on finite clips and suffer from memory explosion, VAM supports online indexing and agentic retrieval over effectively unbounded video through adaptive temporal-visual compression, hierarchical memory, and parallel temporal-spatial representations.

1.2 Project Goal and Scope

Within this problem setting, the project objective is to build and evaluate an online long video understanding assistant for video sources, including egocentric lifelog clips and fixed indoor cameras. The scope prioritises evidence retrieval, temporal grounding, and end-to-end operational efficiency rather than dense caption generation for every frame. Concretely, the system is designed to perform *online indexing* for adaptive evidence preservation across temporal and spatial dimensions, maintain *hierarchical memory* for long-term storage, and utilise *agentic retrieval* to identify the evidence needed to answer user queries. In summary, the motivation for this research stems from the inherent tension between the massive scale of continuous video streams and the limited working memory of current multimodal models. By formalising online long video understanding as a retrieval-driven reasoning problem, this work shifts the paradigm from passive video logging to active, queryable visual memory, establishing the necessary architectural foundation to solve the localisation and hallucination bottlenecks in real-world deployment.

The remainder of this work is organised as follows. Chapter 2 reviews prior methods, benchmarks, applications, and open challenges. Chapter 3 presents the VAM framework, including online indexing, hierarchical memory, agentic retrieval, and asynchronous orchestration. Chapter 4 evaluates the system under both strict online and long-horizon settings. Finally, Chapter 5 discusses implications, limitations, and future directions. Implementation details and prompt-level control contracts are provided in Appendix A and Appendix B.

1.3 Contributions

Despite progress in video retrieval and tool-using agents, building an integrated system that remains reliable over long timelines requires careful orchestration of memory construction and retrieval. The key contributions of this project are:

1. **Adaptive Online Indexing:** A memory construction pipeline for boundary-sensitive moment detection under strict online streaming constraints. By combining heuristic quality gates with an adaptive thresholding mechanism (τ_{dedup}) derived from Otsu partitioning [45], the system suppresses redundancy without future-frame look-ahead.
2. **Hierarchical Memory with Parallel Representation:** A structured memory architecture that organises moments and events across temporal tiers. By maintaining a parallel representation that aligns temporal context with spatial evidence, the system balances scalable storage with the preservation of recoverable visual details.
3. **Agentic Retrieval for Grounded Reasoning:** A multi-turn retrieval loop in which an MLLM acts as a retrieval agent to search, inspect, and summarise evidence. This design enables the decomposition of complex queries into executable retrieval steps, allowing temporal cues and direct visual inspection to be combined for auditable reasoning over distant events.
4. **Training-free End-to-end Framework:** A framework that integrates online indexing, hierarchical storage, and agentic retrieval into a unified system without task-specific retraining. The resulting architecture supports streaming perception and long-horizon retrieval using pre-trained multimodal components.

2 Background and Related Work

Online long video understanding differs fundamentally from short-clip video reasoning because it must operate under two coupled constraints: continuous memory updates under streaming input, and reliable retrieval over long temporal horizons. As video duration extends from minutes to hours, days, or months, relevant evidence becomes sparse, repeated events become harder to disambiguate, and retrieval errors propagate more severely into downstream reasoning [21, 22, 38]. Consequently, the problem is not well characterised by larger context windows alone, but by how prior systems balance online processing, memory organisation, temporal grounding, and long-horizon search [18, 19, 31]. Under the terminology adopted in this work, these pressures map most directly onto online indexing, hierarchical memory, and agentic retrieval as the three core system requirements. The discussion in this chapter is therefore organised through four complementary perspectives: **Methods**, which examine how prior systems structure processing and memory; **Benchmarks**, which clarify what capabilities current evaluations actually measure; **Applications**, which reveal the practical settings that motivate persistent visual memory; and **Challenges**, which isolate the remaining technical gaps that continue to limit reliable long-horizon deployment.

2.1 Methods

Video understanding has evolved from image-level question answering [46, 47] to closed-set supervised video prediction and eventually to open-ended multimodal generation. Early paradigms formalised clip-level settings with constrained answer spaces [23, 24, 48]. Representative models relied on attribute-guided attention, co-memory fusion, and graph-temporal reasoning for question-conditioned evidence aggregation [49, 50, 51]. More recent progress has been driven by increasingly strong video and vision-language backbones, including VideoBERT, Frozen-in-Time, ViViT, PySlowFast, and the InternVideo family, which improved transferable spatiotemporal representations and video-language alignment [52, 53, 54, 55, 56, 57]. In parallel, the field has shifted toward open-form generation with pretrained Multimodal Large Language Models (MLLMs) and language

decoders [40, 41]. This transition is now reinforced by general multimodal model lines that support video as a native input modality, including Gemini 2.5, Gemini 3, Qwen3-VL, and Qwen3.5 [42, 43, 58, 59]. At the same time, video-specialised systems such as VideoChatGPT and InternVideo2-Chat continue to demonstrate the value of stronger video-centric modelling for temporal reasoning, dialogue, and long-context video understanding [57, 60]. In this literature, an embedding model refers to a representation model that maps visual observations or textual queries into a comparable vector space, so that similarity can be computed efficiently for indexing and retrieval [53, 61]. Its role is therefore distinct from that of an MLLM: the embedding model provides the retrieval substrate, whereas the MLLM is typically used for higher-level interpretation, generation, or tool-mediated reasoning. More broadly, these developments show that the field is no longer limited by basic multimodal perception alone; the more difficult question is how to retain, organise, and retrieve visual evidence once video duration extends beyond the short-clip regime.

As summarised in Figure 2.1, current systems can be organised into three macro-paradigms: direct generative inference, compressed memory modelling, and agentic reasoning. Together, these paradigms capture the dominant architectural choices in existing long-video systems: direct end-to-end answering from bounded context, compact memory construction through compression, and iterative planning over preserved or abstracted evidence. This distinction is important because the central trade-off is not merely between model size and context length, but between latency, compression, evidence preservation, and retrieval flexibility.

2.1.1 Direct Generative Inference

Methods based on direct generative inference decode answers directly from in-context visual tokens [60, 62, 63]. As illustrated in Figure 2.1, these approaches occupy the direct-decoding end of the design spectrum and inherit the broader transition from short-clip video QA toward open-form multimodal generation. Early systems were typically built around bounded clips and constrained answer spaces, so the main challenge was local temporal reasoning within a limited window rather than persistent memory over long durations [23, 24, 48]. More recent progress has been driven by increasingly capable MLLMs, where video is no longer treated only as an auxiliary extension of image-language

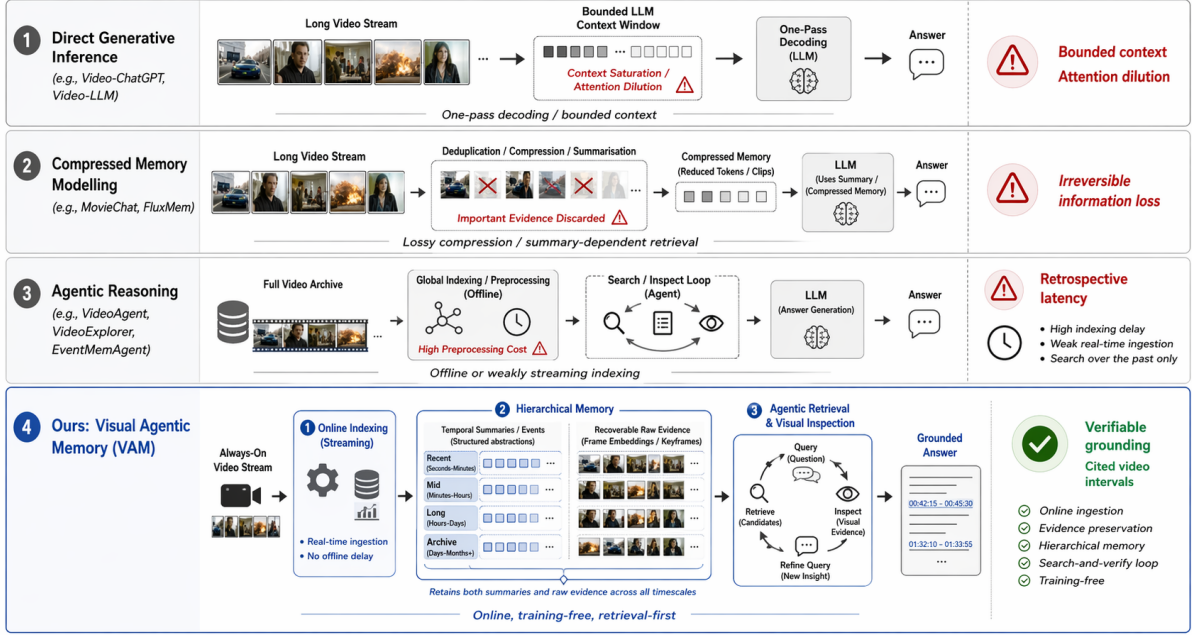


Figure 2.1: Comparison of online long video understanding paradigms. The figure highlights three dominant macro-paradigms in existing systems: direct generative inference, compressed memory modelling, and agentic reasoning. VAM occupies a retrieval-first design point that combines online indexing with explicit agentic retrieval over effectively unbounded video streams.

models but is increasingly supported as a native multimodal input. This trend is visible in both video-specialised systems such as Video-ChatGPT and InternVideo2-Chat [57, 60], and in general multimodal model lines such as Gemini 2.5, Gemini 3, Qwen3-VL, and Qwen3.5, which explicitly broaden native support for video understanding, long-context multimodal interaction, and instruction-following generation [42, 43, 58, 59]. At the same time, current long-video systems still do not exploit these models as fully as they could: in many cases, native video-capable MLLMs are still wrapped inside fixed-rate sampling pipelines or bounded-context prompting schemes, so their improved multimodal interfaces do not by themselves resolve long-horizon memory, retrieval, and grounding. From this perspective, direct generative inference should be understood not as an outdated baseline, but as an increasingly strong default paradigm whose current practical use still falls short of the native capabilities now offered by leading MLLMs.

Its principal weakness, however, remains structural rather than purely representational. Even when the underlying MLLM is substantially stronger, the paradigm still assumes that the task-relevant evidence can be packed into a bounded prompt and resolved within a single decoding pass. In practice, this usually requires fixed-rate sampling, for example

uniform sampling of N frames or one frame per second, which either ignores the semantic density of the stream or leads to rapid context saturation as duration increases. As the temporal horizon expands, these methods become increasingly vulnerable to attention dilution, noise accumulation, and the loss of sparse but decisive evidence [26]. They may produce fluent answers for short or moderately long clips, but they lack a dedicated memory mechanism for preserving and revisiting earlier observations. Consequently, when relevant evidence is buried far back in the stream, appears only briefly, or must be localised precisely rather than summarised impressionistically, direct generative systems become unreliable. This limitation is especially severe in online long video understanding, where the central difficulty is not only multimodal perception, but also the need to retain, retrieve, and verify evidence over hours, days, or months of continuous visual experience.

Closely related streaming baselines refine this paradigm without fully departing from it. SimpleStream shows that a sliding-window policy over only the most recent frames can already match or surpass several published streaming systems, thereby establishing a strong recent-context baseline that more elaborate memory mechanisms must exceed under matched protocols [64]. ViSpeak addresses a different aspect of streaming understanding, namely interaction, by fine-tuning an omni-modal model for visual instruction feedback so that the system can react to gestures, interruptions, and visually expressed user intent in real time [65]. StreamAgent moves further toward proactive control by combining task-driven anticipation with a streaming KV-cache memory, enabling the model to predict which future intervals and spatial regions are likely to contain task-relevant evidence [66]. These systems are important comparators because they strengthen online responsiveness, interaction, or recent-context utilisation without fully adopting a persistent evidence-preserving retrieval architecture. They therefore clarify an important boundary condition for the present dissertation: strong online perception and local streaming competence do not, by themselves, resolve the longer-horizon problems of evidence recoverability, retrospective localisation, and auditable memory over extended streams.

2.1.2 Compressed Memory Modelling

Methods based on compressed memory modelling manage context growth by retaining compact explicit memory instead of preserving all recoverable evidence. This line

emerged as a response to the limitations of direct prompt-based inference: rather than carrying all frames forward, the system builds a condensed intermediate memory that is cheaper to store and query. The core idea is shared across several recent systems, but the concrete mechanisms differ substantially. MovieChat and MovieChat+ rely on sparse retention and memory summarisation to keep only a small set of representative content over time [20, 38]. StreamForest pushes this direction further by treating long streams as a progressively compressed memory hierarchy rather than a flat prompt, so retrieval can operate over summaries before falling back to finer evidence [29]. Infinity-Video uses continuous-time memory consolidation to maintain a rolling compressed account of long videos, aiming to preserve broad temporal coherence without keeping all frame-level detail [67]. FluxMem represents a more aggressive compression-oriented design: it explicitly removes redundancy before storage through adaptive hierarchical memory construction, using mechanisms such as temporal adjacency selection and spatial domain consolidation to compress both temporal repetition and spatial redundancy [28]. Importantly, FluxMem also derives its adaptive token-reduction thresholds from Otsu’s method in both temporal adjacency selection and spatial domain consolidation, rather than relying only on fixed pruning ratios or manually tuned similarity cut-offs [28]. Related online systems such as StreamMind likewise depend on event formation and compressed intermediate memory to remain efficient under streaming input, rather than preserving a dense retrievable trace of the raw stream [36, 68].

What unifies these methods is not a single implementation detail but a common systems choice: they trade evidence fidelity for bounded storage and cheaper downstream processing. Their strengths are clear. Compression-heavy designs can extend the usable temporal horizon substantially, reduce inference cost, and make long videos tractable without requiring fully agentic search over all observations. However, their weaknesses are equally systematic. Sparse retention may under-cover brief state transitions or low-frequency events when query evidence is temporally narrow. Hierarchical or recurrent consolidation can erase visually subtle cues that later become essential for attribution, object disambiguation, or temporal localisation. Boundary-sensitive systems are also exposed to upstream segmentation errors, so an incorrect event split or merge can become a persistent retrieval bias in all later stages. More generally, once fine-grained visual details are discarded during compression, later reasoning may no longer recover the original evi-

dence. The resulting failure modes include weaker citation fidelity, reduced retrospective inspection, and an over-reliance on summary quality rather than direct visual verification [28, 36, 68]. Thus, although compressed memory modelling is substantially more scalable than direct one-pass decoding, many instances are still designed around offline batch processing, static consolidation policies, or lossy online compression rather than persistent searchable memory that remains faithful to the underlying stream.

2.1.3 Agentic Reasoning

Methods based on agentic reasoning interleave planning, search, inspection, or recursive evidence refinement through iterative tool use [3, 30, 31, 34, 35, 69]. This line of work reflects a broader transition in long-video understanding: rather than assuming that a single forward pass over a bounded prompt is sufficient, the model is treated as a controller that decomposes the task into a sequence of retrieval, verification, and synthesis operations. Once video duration exceeds the effective context horizon of one-pass MLLMs, performance depends increasingly on whether the system can localise, inspect, and revise evidence over time. Nevertheless, the agentic literature is methodologically heterogeneous. Existing systems place the principal burden on different substrates, including coarse-to-fine retrieval over stored evidence, graph-structured semantic memory, learned reasoning trajectories, and recursively updated intermediate states. Critically, most prior systems are strongest on one side of the problem more than the other: some are designed primarily for strict online interaction but remain shallow in long-horizon recoverability, while others support longer-range reasoning yet rely on offline or weakly streaming assumptions.

A first subgroup is retrieval-first agentic search, represented by systems such as VideoAgent, EGAgent, LVAgent, VideoExplorer, and EventMemAgent [30, 31, 33, 34, 35]. These methods explicitly search over stored evidence, often combining coarse retrieval, temporal filtering, visual inspection, graph traversal, or iterative reranking. Their principal contribution is to render reasoning procedurally inspectable: the model may formulate hypotheses, request additional evidence, and revise its answer after observing retrieved candidates. This substantially improves traceability and compositionality relative to direct generation, particularly for multi-hop questions or tasks requiring explicit localisation. However, the representational assumptions differ markedly across methods. EGA-

gent centres its reasoning process on entity scene graphs augmented with visual and audio search tools, so the dominant abstraction is a structured semantic graph rather than a persistent raw-evidence memory [31]. LVAgent places greater emphasis on dynamic multi-agent collaboration, where different MLLM agents are selected and refined through discussion and reflection; the principal innovation therefore lies in collaborative query-time reasoning rather than in an always-on memory architecture [70]. VideoExplorer departs from retrieve-then-answer pipelines by advocating “thinking with video”, namely iterative sub-question generation and task-oriented perception directly over video, but does so through an explicitly trained reasoning policy rather than through a training-free persistent memory design [30]. EventMemAgent is closer to streaming settings because it introduces hierarchical event-centric memory and adaptive tool use, yet its primary representational object remains the event abstraction itself rather than a synchronised pairing of raw RGB evidence with temporal memory [35].

This distinction is analytically important. The presence of tool use or visual inspection does not, by itself, imply evidence preservation in the stronger sense required for long-horizon auditable memory. Several retrieval-first agents can revisit selected frames or clips, but they do not generally organise recoverable raw RGB observations as a first-class, persistent, and queryable memory substrate. In many instances, the dominant object of reasoning remains a graph, a caption inventory, an event summary, or another higher-level proxy over the stream. Consequently, retrieval-first agents remain exposed to global localisation collapse: as the searchable memory pool expands linearly with video duration, the search space becomes increasingly noisy, and nearest-neighbour retrieval may fail to distinguish repeated routines or visually similar events [3]. Multi-round controllers can recover difficult cases through iterative expansion and re-checking, but they are also sensitive to early retrieval errors, susceptible to candidate explosion, and vulnerable to rapid latency accumulation as tool calls increase.

A second subgroup places greater emphasis on recursive agentic reasoning over compact intermediate memory, as in ReMA and Temporal Chain of Thought [3, 69]. Here the model not only selects evidence but also progressively updates an increasingly abstract running account of the stream, allowing reasoning to proceed without preserving a dense retrievable trace of the original observations. This design can improve efficiency and

reduce the immediate burden of exhaustive search, but it also transfers greater responsibility to the quality of the intermediate reasoning process itself. If an incorrect hypothesis is formed early, later stages may inherit and reinforce the error rather than correcting it through direct evidence lookup. ReMA makes this trade-off particularly explicit: it addresses the working-memory bottleneck of end-to-end MLLMs through recursive multimodal memory management, but in doing so it moves the system closer to an evolving belief state than to a persistent evidence store [3]. Temporal Chain of Thought similarly improves long-video reasoning by structuring temporal inference more carefully, yet the principal advance remains the reasoning procedure rather than the preservation of a searchable raw-evidence substrate [69]. Such methods are therefore less reliable when the task depends on brief visual details, precise temporal attribution, or retrospective verification against the original observations.

Overall, the agentic family is more auditable and flexible than direct generation, yet it remains difficult to scale gracefully under long timelines and online constraints because both retrieval-first and recursive variants introduce their own forms of latency accumulation, error propagation, and dependence on high-quality intermediate control. The central trade-off is consistent across these paradigms: stronger abstraction improves tractability, whereas fine-grained recoverability and citation fidelity depend on how much original evidence remains accessible at retrieval time. In practice, this yields three recurrent system-level limitations: windowed or selectively searched stores may miss distant but relevant events; abstraction-heavy memory may conceal raw detail that later becomes essential; and iterative agents may violate latency requirements in interactive settings. A scalable design therefore requires a tighter coupling between online indexing and agentic retrieval than is typically provided by retrieval policy alone.

VAM is intended to address this gap. It performs online indexing to maintain a structured, searchable memory of the stream while preserving the conditions necessary for agentic retrieval over hours of visual data. Unlike compression-heavy methods, it is explicitly evidence-preserving; unlike predominantly offline agentic methods, it is designed for continuous and incremental memory updates. More specifically, VAM is designed to satisfy both sides of the problem simultaneously: it remains online under strict no-future-frame constraints, and it remains long-horizon in the sense of maintaining retrievable

evidence over extended timelines. This distinction is especially relevant to world-model-style reasoning [7, 10]. A compact world model summarises the stream into an evolving internal state, but in online long video understanding, excessive abstraction can obscure the very evidence required to determine what the world contained at a particular moment. VAM is therefore more appropriately characterised as an evidence-preserving substrate: its parallel representation keeps the temporal representation aligned with the spatial representation and preserves raw RGB evidence for later visual inspection. Online indexing thus maintains a high-level account of the stream, while agentic retrieval retains the ability to return to the original observations when retrospective inspection is required. In this sense, VAM complements world-model-style reasoning by externalising long-term perceptual memory, allowing higher-level reasoning to operate over retrieved evidence rather than over an irrevocably compressed representation alone.

2.2 Benchmarks

Current benchmarks for long video understanding fall broadly into two families: *online* benchmarks, which test timestamp-conditioned understanding under sequential input, and *long-horizon* benchmarks, which test retrieval and reasoning over extended temporal context without necessarily imposing online interaction constraints. Earlier benchmarks such as MovieQA and TVQA mainly evaluated short-clip understanding and local question answering under relatively bounded temporal scope [24, 48]. More recent benchmark coverage extends to settings such as StreamingBench, LongVideoBench, MLVU, Ego-LifeQA, Video-MME, OVO-Bench, and MM-Lifelong, which collectively broaden the field across streaming interaction, long-context multimodal retrieval, multi-task long-video evaluation, assistance-oriented egocentric memory, and persistent memory assessment [2, 3, 18, 19, 21, 22, 71].

In this work, the empirical evaluation is concentrated on two settings within this broader benchmark landscape: *OVO-Bench* and *MM-Lifelong train@month*.

The first of these, *OVO-Bench*, is the most directly relevant online benchmark because it evaluates strict no-future-frame operation under timestamp-conditioned queries [19]. In the current official benchmark edition used in this work, it contains 644 videos and 3100 queries, with source videos ranging from a few minutes to approximately half an

hour. This makes it a close match to the intended operating regime of online indexing and temporally grounded retrieval.

The second benchmark defines the long-horizon side of the evaluation. *MM-Lifelong* is designed around persistent daily-life memory rather than clip length alone. It is explicitly structured across Day, Week, and Month scales over 181.1 hours of video and 1289 questions [3]. In the month-scale Live Stream portion used here, the data span 105.6 hours across 51 days and contribute 266 *train@month* questions together with 623 *val@month* questions.

These benchmarks also differ in the specific failures they are best suited to reveal. Within the *online* group, StreamingBench is useful for diagnosing failures in streaming perception, cross-modal synchronisation, and contextual interaction under continuous input, while OVO-Bench isolates temporal-awareness failures more directly through backward tracing, real-time understanding, and forward active responding. This makes OVO-Bench especially suitable as the primary benchmark in the present study, because it tests whether a system can maintain usable online memory and recover past evidence under timestamp-conditioned queries rather than merely answer after full-video access. Within the *long-horizon* group, LongVideoBench emphasizes long-context multimodal retrieval and referring reasoning over interleaved video and subtitle evidence, whereas MLVU emphasizes broader long-video capability coverage across genres and task formats. *Video-MME* plays a somewhat different role: rather than targeting only one failure mode, it provides broad multimodal capability coverage across domains, durations, and modality settings, making it useful as a general long-video benchmark. EgoLifeQA and MM-Lifelong become especially relevant when the objective shifts toward persistent perceptual memory under naturalistic daily life. EgoLifeQA stresses assistance-oriented question answering over egocentric experience spanning hours and days, whereas MM-Lifelong explicitly exposes failure modes such as the working memory bottleneck of end-to-end MLLMs and global localisation collapse in agentic baselines over Day, Week, and Month scales [2, 3]. A further dimension is retrieval recoverability, namely whether evidence remains accessible without lossy abstraction. Current benchmarks expose this only partially, typically through downstream temporal attribution, localisation, or answer quality, so systems may score well on end-task responses while still failing near-lossless

evidence recovery in practice.

For this reason, benchmark results should be interpreted along several dimensions rather than as answer quality alone. Particularly important considerations include answer quality, temporal localisation fidelity, retrieval faithfulness or recoverability, and efficiency under continuous streams. Reported gains therefore reflect not only representation quality, but also retrieval policy and temporal inspection behaviour. High answer scores can coexist with poor timestamp alignment, so apparent improvements may still conceal failures in temporal grounding. Benchmarks with denser motion or richer subtitles may likewise overestimate performance for retrieval pipelines that depend heavily on text channels. More fundamentally, when the evaluation protocol does not enforce strict online constraints, methods with retrospective access can appear stronger than systems that operate under genuinely online conditions.

2.3 Applications

While benchmarks provide standardized evaluation, the practical requirements for online long video understanding are rooted in diverse real-world use cases. Egocentric lifelogging remains a primary stress-test scenario because streams are long, untrimmed, and interaction is naturally real-time [1, 2]. From SenseCam to Project Aria and Project Astra, the field has progressively shifted from retrospective memory aids toward continuous multimodal assistance [72, 73, 74]. Recent wearable assistance systems such as Memoro and ProMemAssist further reinforce this shift by emphasizing real-time memory augmentation, proactive support, and context-sensitive assistance rather than post hoc review alone [75, 76]. The same trajectory is now visible in consumer-facing multimodal assistants, including wearable products such as Ray-Ban Meta and Looki, as well as portable AI companions such as rabbit r1, where always-available multimodal perception is increasingly framed as a practical personal-assistance capability rather than only a research benchmark setting [77, 78, 79].

Comparable systems requirements also arise in non-egocentric long-form archives, including surveillance footage, meeting recordings, and personal media repositories, where sparse evidence must be localised within extended recordings and later inspected against the original source material. Across these settings, the technical challenge is not merely

recognising what appears in the stream, but maintaining sufficient temporal structure and evidential recoverability for later search, attribution, and verification.

Application-level defects further clarify design requirements. In egocentric streams, view-point instability and repetitive routines increase false temporal matches; in surveillance, long idle periods create extreme sparsity and low event density; in meetings, semantically similar utterances across agenda segments create cross-turn ambiguity and citation drift. Across all scenarios, failure is rarely binary correctness failure alone; more often it is locatability failure, attribution failure, or inspection failure. These failure classes motivate system designs that optimise not only answer quality, but also searchability, temporal precision, and retrospective auditability.

Overall, the application literature suggests that practical deployment requires a transition from benchmark-level competence to system-level reliability. Addressing sparsity, repetitiveness, and inspection failures is therefore not an auxiliary concern, but a prerequisite for continuous multimodal assistance.

2.4 Challenges

Despite rapid progress, the central difficulty remains split across two requirements that are often addressed separately rather than together. The first is an *online challenge*. Under streaming input, the system must decide what to retain, merge, or discard while relying only on observations available so far and still meeting bounded-latency response [18, 19]. However, these same choices can remove brief but decisive events before they ever become queryable.

The second is a *long-horizon challenge*. Over hours, days, or months, the system must not only scale, but remain searchable, temporally precise, and visually recoverable. Here the dominant failure mode is not immediate online responsiveness, but recoverability collapse: a flat retrieval pool becomes noisy and ambiguous as repeated events accumulate, while a heavily compressed memory may preserve semantic gist only by discarding the original visual evidence needed for later verification. In many systems, the raw images themselves cease to be available as first-class evidence, leaving only summaries, captions, event abstractions, or other consolidated proxies [3, 31, 44]. This weakens later inspection precisely when fine-grained object state, spatial detail, or photometric evidence matters

most.

The core trade-off is that many prior systems effectively optimise one side of the problem by weakening the other. Systems tuned for online operation often compress aggressively, keep only shallow recent memory, or sacrifice long-range recoverability to remain efficient. Systems tuned for long-range reasoning often support richer search or iterative verification, yet rely on offline preprocessing, weak streaming assumptions, or memory structures that are too expensive to update continuously. Query answering in online long video understanding therefore becomes an *orchestration challenge* between memory construction and reasoning: the system must preserve evidence online, organise it over long horizons, and still support search, Visual Inspection, and justification at query time.

Online indexing addresses the strict online requirement by updating memory incrementally under streaming constraints. Hierarchical memory with a parallel representation addresses the long-horizon requirement by keeping temporal event structure and frame-level evidence in the spatial representation jointly available with the temporal representation, rather than forcing a choice between searchable summaries and recoverable raw observations. Agentic retrieval, coupled with asynchronous orchestration, addresses the coordination problem by allowing search, visual inspection, and justification to operate over the same shared memory without blocking continuous indexing.

From a world-understanding perspective [5,6,7,11], this integration matters because the deeper issue is not memory scale alone, but whether a system can maintain an interpretable account of the world without giving up access to the observations from which that account was formed. Under this view, VAM can serve as a persistent perceptual memory layer for future world-model systems: it keeps the stream searchable at fine granularity, then allows downstream reasoning to assemble and inspect the task-relevant slice of the world at query time. The design objective is therefore not only correct answering, but auditable world-state understanding under latency and storage constraints.

3 Methodology

The VAM framework is formulated as a method for transforming a continuous video stream into a searchable memory trace that remains available for later retrieval, inspection, and citation. As frames arrive, the system filters and indexes the stream online, writes retained observations into hierarchical memory, and answers user requests through retrieval and visual inspection over the resulting memory. The methodological objective is to retain recoverable visual evidence while keeping long-horizon search tractable, so that later reasoning remains grounded in observations that can still be revisited directly rather than only in compressed hidden state or summary-only abstractions. Satisfying this objective requires continuous memory construction to remain separate from downstream reasoning, such that online indexing proceeds as the stream evolves while long-term storage and query-time inspection remain coordinated with, but are not blocked by, multi-turn retrieval decisions.

3.1 System Architecture

At a system level, the framework consists of three coupled components: an online indexing process (Section 3.2), a hierarchical memory substrate (Section 3.3), and a query-time agentic retrieval process (Section 3.4). The indexing process updates memory incrementally as new frames are processed, with each indexing decision made without reliance on future observations. Incoming video is first processed by online indexing, where low-quality and redundant frames are filtered and visually significant transitions are retained. These retained transition points are written as moments, which form the frame-level units that preserve raw visual observations together with spatial embeddings. Contiguous moments are then grouped into events, which provide temporally extended units associated with event documents and temporal metadata. This organisation yields a parallel representation in which temporal documents support coarse retrieval while the underlying spatial evidence remains available for direct visual inspection.

At query time, agentic retrieval transforms natural-language requests into executable

search plans. An MLLM, acting as the retrieval agent, iteratively consults temporal and spatial retrieval pathways to identify candidate evidence segments, request additional inspection when necessary, and synthesise answers against recoverable visual intervals. The same multimodal model family is also used earlier in the pipeline for event summarisation, so summarisation and inspection are treated here as two operational roles of a shared MLLM interface rather than as unrelated module classes. Figure 3.1 provides a high-level view of this workflow and the resulting interaction between online indexing, hierarchical memory, and query-time retrieval.

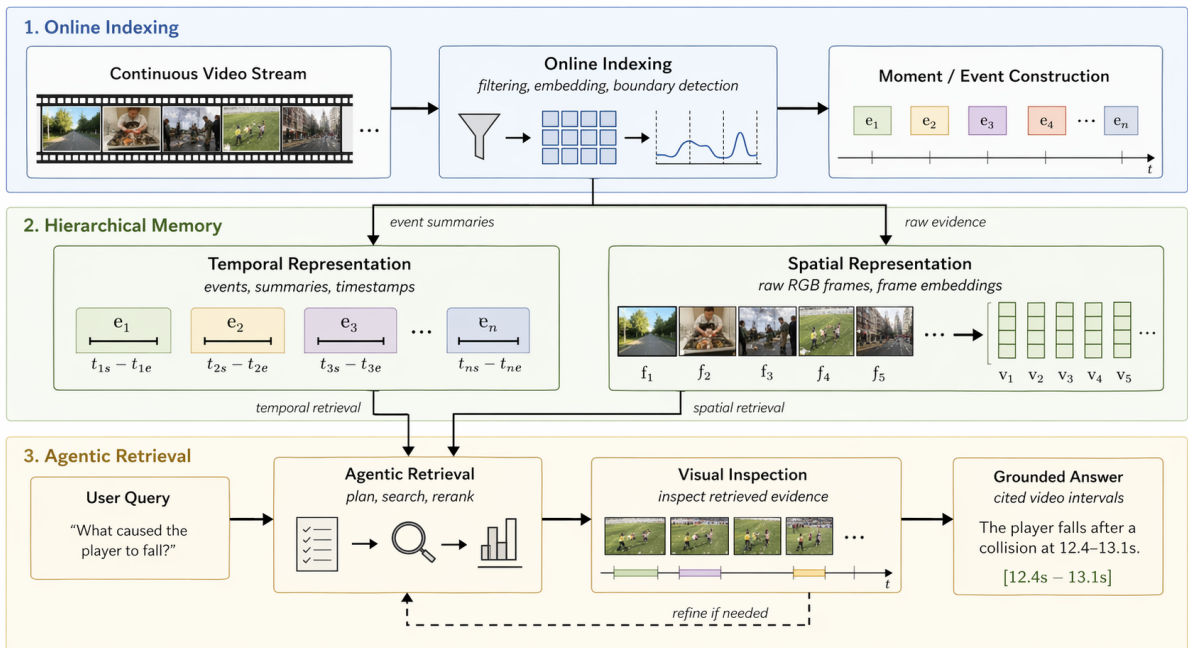


Figure 3.1: System architecture of VAM. The framework is organised into three coupled layers: Online Indexing, Hierarchical Memory, and Agentic Retrieval. In the indexing layer, the continuous video stream is filtered, embedded, and segmented through boundary-sensitive updates, producing retained moments that are subsequently grouped into events. These outputs are written into a parallel memory structure with two synchronised views: a temporal representation that stores events, event documents, timestamps, and optionally materialised summary documents for coarse timeline-level retrieval, and a spatial representation that preserves raw RGB evidence together with frame embeddings for fine-grained visual inspection. At query time, Agentic Retrieval reads from both representations, combines temporal and spatial retrieval signals to identify candidate evidence, invokes visual inspection when disambiguation is required, and returns a grounded answer supported by cited video intervals. The dashed feedback link indicates that inspection outcomes can trigger additional retrieval steps before final answer synthesis.

3.2 Online Indexing

Online Indexing is the entry-stage process that allows VAM to remain responsive under continuous streaming input. Its role is not merely to reduce computation, but to decide which visual observations should be retained as future evidence and which can be safely discarded as blur or redundancy. This step is therefore the first place where the system balances efficiency against recoverability: if the filter is too weak, storage and retrieval become noisy; if it is too aggressive, later retrieval may lose the very frames needed for visual inspection.

Concretely, consider an incoming stream $\mathcal{F} = \{F_1, F_2, \dots, F_N\}$, where each frame F_i is associated with both a relative timestamp t_i and an absolute timestamp T_i . The goal of online indexing is to transform this dense raw stream into a much sparser yet still evidence-preserving memory trace through a sequence of filtering, embedding, and boundary-update operations before higher-level temporal encoding. At a high level, this pipeline performs three consecutive functions: it first removes observations that are unlikely to support later reasoning, then converts the retained frames into a representation suitable for retrieval, and finally decides where memory boundaries should be written so that long streams remain searchable without losing critical transitions. The retained frame-level entries written by this process are referred to as *moments*, where each moment stores a retained frame together with its embedding and timestamps; temporally contiguous groups of moments later define higher-level *events*.

Stage 1: Frame Filtering

The first stage addresses a basic but consequential problem in continuous video streams: not every incoming frame is worth preserving or even encoding. In real deployments, cameras frequently produce motion blur, viewpoint jitter, and near-duplicate observations during quasi-static periods. If all such frames were forwarded downstream, the system would spend compute on visually uninformative content and would also contaminate the memory store with repeated evidence. Stage 1 therefore acts as a protective gate whose purpose is to improve both efficiency and memory quality before any higher-cost neural processing is applied.

In practice, the filtering stage removes observations that are either visually unreliable or

insufficiently novel. The first criterion is image quality, and the second is pre-embedding redundancy. Only frames that survive both checks are allowed to enter the representation pipeline.

Blur Detection (Laplacian Variance): The first gate targets severe blur, because heavily blurred frames rarely provide trustworthy object, scene, or state information for downstream retrieval. We therefore estimate frame sharpness using the variance of the Laplacian operator:

$$\text{Sharpness}(I_t) = \text{Var}(\nabla^2 I_{t,\text{gray}}) \quad (3.1)$$

The Laplacian operator highlights regions of rapid intensity change. A blurry image yields lower variance; if $\text{Sharpness}(I_t) < \tau_{\text{blur}}$, the frame is discarded.

Redundancy Filtering: Blur removal alone is insufficient, because many streams remain dominated by visually near-identical frames even when image quality is acceptable. To avoid repeatedly encoding content that has not meaningfully changed, the system performs pre-embedding redundancy suppression based on visual similarity. This is deliberately placed before dense embedding extraction so that obviously repeated content can be discarded as early as possible. Frames judged visually unchanged are removed, and only those passing both the sharpness and redundancy gates proceed to the embedding stage. Figure 3.2 provides a qualitative example of this filtering stage, showing how blurred or near-duplicate frames are removed before embedding extraction. Detailed gate ordering and default values are provided in Appendix A.

Stage 2: Embedding Extraction

After filtering, the system still requires a representation that supports efficient comparison across time. Raw frames preserve the underlying evidence, but they are not by themselves a practical index for large-scale retrieval, boundary detection, or similarity-based matching. Stage 2 therefore converts each retained frame into a dense feature vector that captures its semantic and visual content in a form that can be compared quickly and consistently.

For each frame passing the quality gates, the system extracts a dense embedding $\mathbf{e}_t \in \mathbb{R}^d$ using a pre-trained multimodal embedding model [53, 61]. The specific embedding instantiation is treated as an experimental choice rather than as a fixed architectural re-

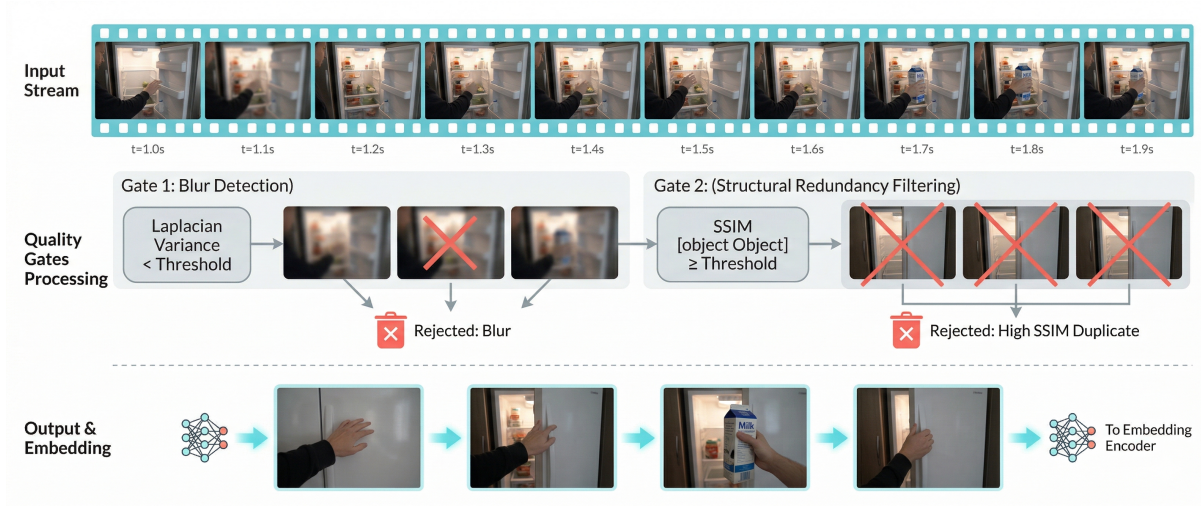


Figure 3.2: Visual illustration of the frame filtering stage (Stage 1). The top row shows an extended sequence of the raw incoming video stream. The middle row highlights frames rejected due to extreme motion blur (Laplacian variance $< \tau_{blur}$) or low-change redundancy during pre-embedding filtering. The bottom row displays the retained keyframes that proceed to the embedding extraction stage.

quirement. Here, the embedding model serves as the retrieval-facing representation layer of the system. Its purpose is not to generate summaries or final answers, but to map retained frames into a shared feature space in which visual similarity can be measured efficiently. This representation supports three downstream functions simultaneously: boundary-sensitive change detection during online indexing, frame-level retrieval over the spatial memory, and alignment between user queries and candidate visual evidence. In this sense, the embedding model provides the searchable substrate of VAM, while the MLLM is reserved for event summarisation, visual inspection, and answer synthesis.

Stage 3: Memory Update and Boundary-Preserving Moment Commit

After frame filtering and embedding extraction, the system must decide whether a newly arrived frame provides genuinely new information or merely repeats what is already represented by the current memory state. This decision is critical because VAM does not aim to store every frame uniformly; instead, it aims to preserve boundary-defining transitions while compacting quasi-static intervals. In other words, the system needs a principled criterion for when a stream has changed enough to justify writing a new memory unit.

To operationalise this decision, the system compares each incoming embedding with the current reference frame in feature space and uses that discrepancy as the signal for memory update:

$$d_c(\mathbf{e}_t, \mathbf{e}_{ref}) = 1 - \frac{\mathbf{e}_t \cdot \mathbf{e}_{ref}}{\|\mathbf{e}_t\| \|\mathbf{e}_{ref}\|} \quad (3.2)$$

However, a single fixed threshold is often brittle under real streaming conditions, because the magnitude of embedding variation depends on scene dynamics, camera motion, and environmental stability. For this reason, VAM does not treat change sensitivity as globally constant. Instead, it maintains a sliding distance history $\mathcal{H}_d$ and derives the adaptive deduplication gate from recent distance statistics, allowing the update rule to adapt to the local temporal regime. Concretely, when sufficient samples are available, the implementation applies Otsu partitioning [45] over $\mathcal{H}_d$; otherwise it falls back to a configured similarity-derived default. This choice is also methodologically close to FluxMem [28], which likewise uses Otsu-derived adaptive thresholds for token reduction, although it applies them within temporal adjacency selection and spatial domain consolidation rather than within an evidence-preserving indexing pipeline:

$$\tau_{dedup} \approx \text{AdaptiveThreshold}(\mathcal{H}_d). \quad (3.3)$$

Frames with $d_c \leq \tau_{dedup}$ are buffered as redundant context. When $d_c > \tau_{dedup}$, the buffered endpoint is committed to memory as a moment and the new frame becomes the reference. In practical terms, quasi-static segments are compacted, while transition points are explicitly retained as frame-level memory entries. This boundary-preserving update rule follows onset/offset retention principles established in memory compression literature [30, 31]. Figure 3.3 visualises this adaptive update mechanism, where distance spikes trigger memory commits and reference resets.

These committed moments do more than control storage density. They also provide the temporal skeleton from which higher-level events are later formed in hierarchical memory. Rather than introducing a separate segmentation heuristic, the implementation reuses the same Otsu-based change analysis at the event level: distances between consecutive retained moments are scanned for local peaks above a relaxed Otsu-derived boundary threshold, and an additional duration cap prevents any single event from becoming arbitrarily long. Event formation therefore remains tied to the same evidence-preserving

change signal used during online indexing.

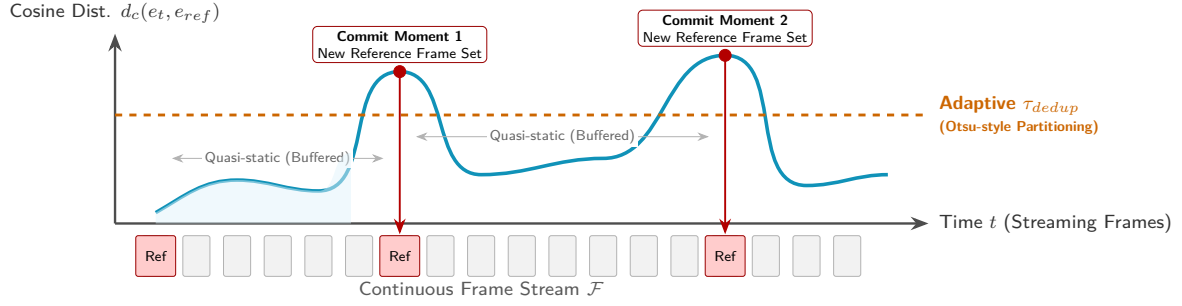


Figure 3.3: Adaptive deduplication and boundary-sensitive updates. The system continuously compares incoming frames to a reference frame. When the visual distance d_c exceeds the adaptive threshold τ_{dedup} , a memory update is triggered, buffered moments are committed, and the reference frame is reset. These distance spikes also provide the primary signal later used for event formation.

At runtime, each incoming frame undergoes four steps: (1) encoding to $\mathbf{e}_t$, (2) computing d_c against the current reference, (3) comparing with the adaptive threshold τ_{dedup} , and (4) either buffering as redundant context or committing a boundary-preserving update. This process is strictly online [30,31], making decisions based on current statistics without requiring future frames.

3.3 Hierarchical Memory

Embeddings alone are insufficient for episodic retrieval over long timelines. Precise frame matching and abstract event recall require complementary indices, as object-level retrieval and timeline reasoning fail under different conditions. Here, a *moment* denotes a retained frame-level memory entry written during online indexing, whereas an *event* denotes a temporally contiguous group of such moments. The system organises these moments into events so that the spatial representation and the temporal representation remain synchronised through shared temporal anchors.

The first organising principle of hierarchical memory is age-aware storage. Retained items are organised into *recent*, *mid*, and *long* tiers according to their age Δt . As events age, compression strength and minimum temporal spacing increase. To preserve recoverability, recent windows retain denser full-frame payloads, while older segments are consolidated into more compact event-level temporal cues that remain retrievable [30,31]. This policy allocates high-fidelity storage to intervals with likely short-term query demand

while ensuring long-term context remains searchable.

The second principle is event formation. In the implementation, event boundaries are detected from the distance sequence between consecutive retained moments. This stage does not introduce a new segmentation signal; instead, it reuses the same Otsu-derived boundary statistics described above, selecting local peaks above a relaxed threshold as split points while also enforcing a maximum event duration. Let m_j denote the j -th retained moment. Consecutive moments between two such boundaries define an event:

$$E_i = \{m_{s_i}, \dots, m_{e_i}\} \quad (3.4)$$

For each event E_i , the system generates an event document from the same temporally bounded visual evidence that defines the event. In the current implementation, the MLLM receives a video clip cropped from the source video over the event interval when available, together with a small set of representative retained frames and temporal metadata; otherwise it falls back to the representative frames and metadata alone:

$$C_i = \text{MLLM}(E_i) \quad (3.5)$$

Here, MLLM again denotes the same multimodal model family used throughout the framework. Online event summarisation and fine-grained query-time inspection are therefore treated as two role-specific uses of that shared multimodal interface rather than as separate module classes.

The generated text C_i is persisted as the event document associated with E_i . Consequently, each event maintains: (i) retained moments, (ii) frame embeddings $\mathbf{e}_t$, (iii) an event-level textual document C_i , and (iv) temporal metadata (t, T) . The spatial representation comprises retained frames (moments) and embeddings, while the temporal representation consists of event documents and time-aware retrieval cues. Together, this parallel representation supports both temporal retrieval and spatial retrieval. Broader *summary documents*, when needed, are generated later at query time by aggregating one or more event documents rather than replacing them.

In simpler terms, this stage supports two different query types within one event: exact visual matching (handled by the spatial representation) and timeline-level reasoning

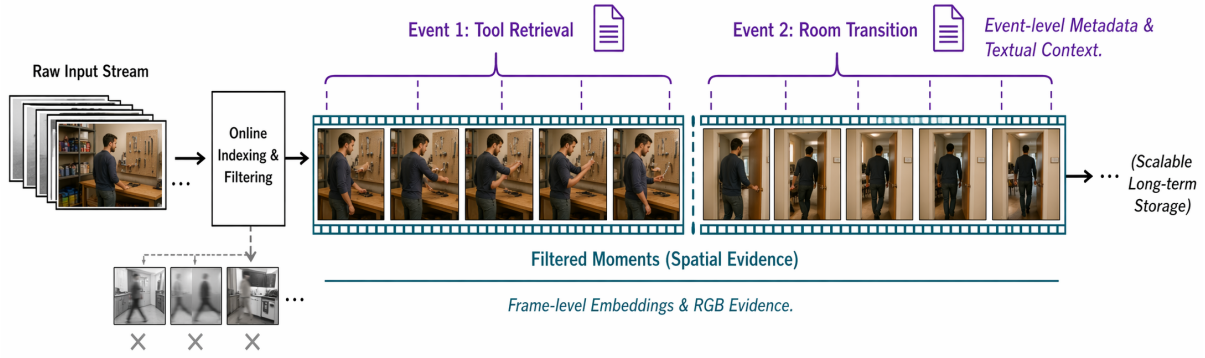


Figure 3.4: Hierarchical memory organisation and the parallel representation. The system filters the raw video stream into a sparse sequence of retained frames (filtered moments), which are then logically grouped into semantic containers (events). Each event maintains a dual representation: a spatial representation consisting of raw frames and frame-level embeddings for fine-grained inspection, and a temporal representation consisting of event documents, temporal metadata, and optionally materialised summary documents for timeline-level search. This structure ensures that reasoning remains grounded in recoverable visual evidence.

(handled by the temporal representation). Keeping both representations synchronised is essential for agentic retrieval, as the agent can transition from event-level temporal cues to frame-level spatial lookup without cross-index reconciliation.

Figure 3.4 shows how retained moments are organised into events and stored through a parallel representation spanning spatial and temporal views. The key architectural point is that frame-level evidence and event-level documents are produced from the same temporally bounded input and remain aligned through shared temporal metadata, so later retrieval can move from event-level temporal hypotheses to frame-level spatial evidence without alignment loss.

In summary, organising incoming moments into these event-level memory units forms the crux of the system’s retrievability. By synchronising frame-level spatial features and temporally anchored event documents under identical metadata, hierarchical memory circumvents the typical trade-off between timeline abstraction and fine-grained visual fidelity, providing a complete basis for subsequent agent queries. This design treats the preservation of raw visual evidence not merely as a storage choice, but as a core requirement for grounded agentic retrieval. Broader summary documents, when needed, are derived later from this substrate rather than replacing it.

3.4 Agentic Retrieval

Agentic Retrieval is formulated as a bounded multi-turn loop that allows the system to bridge the gap between initial coarse-grained hypotheses and fine-grained visual evidence. At each turn k , the MLLM receives the user query Q alongside the accumulated reasoning history $\mathcal{C}_{k-1}$, and selects an action a_k from the defined set:

$$a_k \in \{\text{search}, \text{inspect}, \text{summarize}, \text{answer}\}.$$

This recursive interaction follows the observe-hypothesize-inspect paradigm [30, 31]. To operationalise this loop, the MLLM utilises a set of discrete retrieval primitives, each defined by a formal schema and specific interaction logic with hierarchical memory:

Search (`search`) is the primary operator for broad-scale navigation. It performs a hybrid retrieval step over the temporal and spatial representations, supporting both textual and appearance-based queries. The `inspect_k` parameter enables an integrated look-ahead where top candidates are automatically passed to inspection to accelerate grounding.

Inspect (`inspect`) is the operator used for visual inspection. Unlike the broad `search`, `inspect` performs a deep-dive into specific visual intervals or previously identified candidates. It retrieves raw visual evidence to resolve semantic ambiguities, particularly when precise anchor inspection is required.

Summary Creation (`summarize`) is an on-demand materialisation operator. It synthesises localised event documents and representative frames into a cohesive *summary document*, creating a reusable temporal abstraction that is then persisted in memory.

Answer (`answer`) is the termination primitive. It signals that the agent has gathered sufficient evidence to resolve the user’s query, providing a synthesised response anchored to the `best_ref` for auditable grounding.

Collectively, these primitives define the operational action space through which Agentic Retrieval interacts with hierarchical memory. Rather than allowing free-form generation to proceed without explicit control, the framework constrains inference to a bounded sequence of interpretable retrieval actions, so that hypothesis formation, evidence acquisition, intermediate abstraction, and final response synthesis remain coupled to concrete

memory operations. This design serves two methodological purposes. First, it preserves auditability by ensuring that each reasoning step can be traced to recoverable evidence returned from the temporal or spatial representation. Second, it stabilises long-horizon search by structuring the agent’s behaviour around a small set of reusable operators whose invocation depends on the current evidence state, rather than on unconstrained latent deliberation alone. The four functional operations below are the operational realisation of these primitives within the multi-turn retrieval loop:

Intent Parsing and Search Specification. Each turn begins by converting the natural-language query into an operational search specification. For complex or compositional requests, the MLLM decomposes the user’s intent into a sequence of executable sub-queries, making implicit temporal constraints and entity-level requirements explicit before retrieval is attempted. This stage is particularly important for long-horizon settings, where relevant evidence may be temporally dispersed and cannot be recovered reliably through a single undifferentiated lookup. The result is a structured action specification containing target entities, temporal constraints, operator selection, and source prioritisation.

Hybrid Temporal-Spatial Retrieval. Given this structured specification, the **search** operator consults two complementary retrieval pathways within a single hybrid retrieval process. Let $\mathbf{q}_k$ denote the retrieval embedding derived from the turn- k search specification. Let $D_{TEMP} = \{d_i\}$ denote the set of temporal documents, where each d_i is either an event document or a previously materialised summary document with embedding $\mathbf{v}_{d_i}$. The temporal pathway prioritises broad situational context, whereas the spatial pathway preserves access to fine-grained visual evidence. Used together, they address the central trade-off between timeline abstraction and visual fidelity:

- **Path 1: Temporal Retrieval.** This branch queries event-level documents and summary documents using a RAG-style embedding retriever. Event documents are the localised textual records produced during online indexing, whereas summary documents provide reusable temporal abstractions over broader windows. This distinction is important because standard retrieval over local event documents does not guarantee cross-event consolidation or consistent window-level phrasing.

Summary documents therefore materialise a coarser temporal view by aggregating overlapping evidence into a single retrieval-oriented description:

$$S_{TEMP}(d_i) = \cos(\mathbf{q}_k, \mathbf{v}_{d_i}) \quad (3.6)$$

Temporal retrieval is prioritised for establishing situational context and resolving high-level intent, anchoring the reasoning process within a broad temporal window.

- **Path 2: Spatial Retrieval.** The query is also compared directly against frame-level embeddings:

$$S_{VIS}(F_t) = \cos(\mathbf{q}_k, \mathbf{e}_t) \quad (3.7)$$

Spatial retrieval pinpoints object-level details and short-lived visual events. The agent can also use `visual_ref` for appearance-based retrieval, using a previously identified moment as a visual query to find similar instances.

Grounded Visual Inspection. The candidates returned by retrieval remain provisional hypotheses rather than final evidence. To establish citation fidelity, the MLLM invokes `inspect` to forward the highest-priority candidates for direct examination over the underlying visual evidence. This step may also be executed as a `joint_inspection` during the search turn when the agent must compare multiple temporally related candidates or identify broader temporal patterns. The inspection module evaluates the retrieved evidence as follows:

$$O_k = \text{Inspector}(\mathcal{X}_k^{vis}, \mathcal{X}_k^{temp}, Q) \quad (3.8)$$

where $\mathcal{X}_k^{vis}$ and $\mathcal{X}_k^{temp}$ denote the visual and temporal candidates selected for inspection at turn k . This inspection step resolves semantic ambiguities, adjudicates conflicts between temporal and spatial retrieval signals, and serves as the primary safeguard against unsupported answer generation.

Answer Synthesis and Termination. Termination is implemented as an explicit planner action rather than as an implicit side effect of retrieval. At each turn, the planner may emit `action=answer` once the accumulated context contains sufficiently useful evidence; operationally, this decision is also encouraged when the remaining turn budget

becomes small or when successive turns cease to improve the retrieved result set. The final answer is not generated in an unconstrained manner. Instead, the planner is required to associate the response with a specific supporting reference, denoted `best_ref`, expressed over the turn-indexed retrieval history, typically as a `(turn_idx, result_idx)` pointer to the candidate judged most probative for the query. This reference is then resolved against the stored turn context so that the final response remains coupled to the underlying evidence instance, including its textual description, temporal location, and retrieval score.

This makes answer synthesis explicitly citation-aware. When a valid `best_ref` is provided, the response is grounded in a concrete retrieved result rather than only in the agent’s latent summary of prior turns. When such a reference is absent or invalid, the controller instead falls back to the highest-scoring retrieved candidate accumulated so far, thereby preserving the principle that answer generation should remain anchored to the strongest available evidence. The stopping policy is similarly evidence-first: if search progress stagnates across turns, or if the loop exhausts its maximum turn budget K , the controller forces termination and returns the best available fallback answer derived from the retrieved candidates rather than permitting unbounded exploration. The final output therefore preserves not only the answer text, but also a structured account of whether adequate evidence was found, which candidate was treated as the principal support, the full multi-turn trace, and the total search duration. Termination is thus not merely a procedural stopping rule; it is the stage at which the accumulated retrieval history is consolidated into a directly answerable yet still auditable claim grounded in recoverable visual evidence.

3.5 Asynchronous Orchestration

A central systems requirement of VAM is that continuous memory construction should not require query-time reasoning to control the indexing loop. To satisfy this requirement, the framework adopts an asynchronous orchestration design in which Online Indexing and Agentic Retrieval operate as decoupled yet coordinated processes over a shared hierarchical memory substrate. Online Indexing runs in the background, filtering incoming frames, updating memory, and writing new moments and events as processing contin-

ues; broader summary documents can be materialised later at query time when needed. Agentic Retrieval, by contrast, is instantiated on demand and interacts with memory only through bounded retrieval operators, allowing complex multi-turn reasoning to proceed without taking over the ongoing indexing process [18].

The role of orchestration is therefore not merely to separate two workloads, but to coordinate their interaction under latency and consistency constraints. Query-time reasoning must remain responsive, while the evidence it cites must still be grounded in recoverable memory entries produced by the indexing process. To satisfy these requirements, the framework imposes explicit resource bounds, including turn budgets, candidate set sizes, and inspection batch limits, so that retrieval cost remains predictable even during extended operation. In effect, the orchestration layer turns continuous perception and on-demand reasoning into two concurrently running processes coupled through a common memory interface, thereby preserving both online responsiveness and evidential traceability.

This cost structure also has a practical consequence for empirical evaluation. Because long-horizon retrieval in VAM often requires passing multiple candidate intervals or frame sets to the MLLM for direct inspection, evaluation becomes substantially more expensive than in purely retrieval-only or one-pass decoding settings. As in many recent long-video studies, the empirical scope is further constrained by the absence of unlimited commercial inference budgets and large-scale serving access for repeated multi-image MLLM calls. The study therefore adopts a targeted rather than exhaustive evaluation design, focusing on representative benchmark subsets and targeted evaluation regimes that are most informative for online responsiveness, retrospective evidence recovery, and grounded answer generation, rather than attempting an exhaustive sweep over every available split or benchmark configuration.

4 Evaluation

This chapter evaluates VAM along the two directions that define the dissertation: on-line performance under strict streaming constraints, and long-horizon performance over substantially extended temporal spans. The first direction asks whether VAM functions as a genuinely online system whose memory is updated incrementally and queried using only observations available up to the current timestamp. The second examines retrieval and reasoning over ultra-long video horizons, including month-scale memory settings in which relevant observations may be widely separated in time and difficult to localise. Taken together, these two directions test whether the same memory-and-retrieval architecture remains effective both under strict streaming conditions and under substantially extended temporal horizons.

4.1 Experimental Scope

The evaluation is organised around the two directions introduced above. Online operation is evaluated with *OVO-Bench* [19], whose timestamp-conditioned setup directly tests whether memory can be updated incrementally while each response remains grounded only in observations available up to the query timestamp. Long-horizon retrieval and reasoning are evaluated on the month-scale portion of *MM-Lifelong* [3], which stresses persistent memory over substantially longer real-world temporal spans.

These two benchmark settings differ not only in evaluation protocol, but also in the amount of video, the number of questions, and the temporal extent over which relevant evidence may be distributed. In the current official edition used here, *OVO-Bench* contains 644 videos and 3100 queries, with source videos ranging from a few minutes to approximately half an hour. *MM-Lifelong* contains 181.1 hours of footage and 1289 questions organised across *day*, *week*, and *month* scales, with physical temporal span extending up to two months. The month-scale *Live Stream* subset used here spans 105.6 hours across 51 days and contributes 266 *train@month* questions and 623 *val@month* questions.

Long-horizon evaluation is substantially more expensive than *OVO-Bench* because each query may require large-memory retrieval, repeated visual inspection, and longer agent interaction over temporally distant evidence. The practical scope of evaluation is therefore also shaped by the computational and serving cost of repeated multi-image MLLM calls at extended temporal horizons. In this work, the long-horizon results are reported on *MM-Lifelong train@month*. Following the benchmark protocol of Chen et al. [3], this comparison should be interpreted carefully: end-to-end MLLMs are evaluated by uniform sparse sampling over the stream up to their maximum context capacity, rather than by densely ingesting the full lifelong video; agentic baselines are otherwise run under their official default settings, with model alignment adjustments reported explicitly by the benchmark authors when used for fairness. Across the reported VAM results here, the default configuration uses Gemini Embedding 2 as the embedding backbone for memory indexing and retrieval, Gemini 3 Flash as the MLLM, and an initial frame sampling rate of 0.5 fps; the retained memory itself is then formed adaptively rather than stored at a constant fps.

4.2 OVO-Bench Results

OVO-Bench is used as the primary benchmark for online evaluation because it measures timestamp-conditioned understanding using only the stream prefix available at the query time [19]. This protocol aligns closely with the intended operating regime of VAM, where memory is updated incrementally as the stream unfolds and each query must be answered using only observations available up to the query timestamp. OVO-Bench is therefore relevant not simply as a generic video benchmark, but as a direct test of whether online indexing and agentic retrieval remain effective under strict streaming constraints. On the main OVO-Bench table, VAM achieves the highest score, defined as the mean of the Real-Time Visual Perception and Backward Tracing averages, and this result exceeds that of every compared method.

The benchmark decomposes online performance into three modes. *Real-Time Visual Perception* measures immediate understanding of the current scene, *Backward Tracing* measures whether the system can recover relevant earlier evidence, and *Forward Active Responding* measures whether the system can defer commitment until sufficient evidence

Model	Real-Time Visual Perception							Backward Tracing				Overall
	OCR	ACR	ATR	STU	FPD	OJR	Avg.	EPM	ASI	HLD	Avg.	
Human												
Human Agents	93.96	92.57	94.83	92.70	91.09	94.02	93.20	92.59	93.02	91.37	92.33	92.77
Proprietary MLLMs												
Gemini 3 Flash [59]	<u>86.58</u>	68.81	79.31	58.99	70.30	70.65	72.44	58.92	76.35	52.15	62.47	67.46
GPT-4o [80]	69.80	64.22	71.55	51.12	70.30	59.78	64.46	57.91	<u>75.68</u>	48.66	60.75	62.61
Open-Source Offline MLLMs												
Qwen2-VL-72B [81]	65.77	60.55	69.83	51.69	69.31	54.35	61.92	52.53	60.81	<u>57.53</u>	56.96	59.44
LLaVA-Video-7B [82]	69.13	58.72	68.83	49.44	74.26	59.78	63.36	56.23	57.43	7.53	40.40	51.88
LLaVA-OneVision-7B [83]	66.44	57.80	73.28	53.37	71.29	61.96	64.02	54.21	55.41	21.51	43.71	53.86
Qwen2-VL-7B [81]	60.40	50.46	56.03	47.19	66.34	55.43	55.97	47.81	35.48	56.08	46.46	51.22
InternVL-V2-8B [84]	67.11	60.55	63.79	46.07	68.32	56.52	60.39	48.15	57.43	24.73	43.44	51.92
LongVU-7B [85]	53.69	53.21	62.93	47.75	68.32	59.78	57.61	40.74	59.46	4.84	35.01	46.31
Qwen3-VL-8B [42]	75.17	58.72	72.41	57.30	70.30	59.24	65.52	56.57	69.59	12.37	46.18	55.85
Open-Source Online MLLMs/Agents												
Flash-VStream-7B [86]	24.16	29.36	28.45	33.71	25.74	28.80	28.37	39.06	37.16	5.91	27.38	27.88
VideoLLM-online-8B [87]	8.05	23.85	12.07	14.04	45.54	21.20	20.79	22.22	18.80	12.18	17.73	19.26
Dispider [88]	57.72	49.54	62.07	44.94	61.39	51.63	54.55	48.48	55.41	4.30	36.06	45.31
Streamo-7B (2fps*) [89]	77.18	66.06	76.72	45.51	66.34	72.83	67.44	55.56	58.11	33.87	49.18	58.31
ViSpeak-7B [65]	75.17	58.72	71.55	51.12	74.26	66.85	66.28	<u>59.93</u>	48.65	63.98	57.52	61.90
SimpleStream [64]	94.00	85.30	82.80	<u>65.70</u>	77.20	83.20	81.37	51.90	58.10	52.10	54.03	<u>67.70</u>
FluxMem [28]	81.20	59.60	70.70	53.40	<u>75.20</u>	63.00	67.18	-	-	-	-	-
ThinkStream-3B [36]	85.23	64.22	69.82	49.43	69.31	64.13	67.02	53.87	59.46	43.55	52.29	59.66
StreamForest-7B [29]	68.46	53.21	71.55	47.75	65.35	60.87	61.20	58.92	64.86	32.26	52.01	56.61
StreamAgent [66]	71.20	53.20	63.60	53.90	67.30	58.70	61.32	54.80	58.10	25.80	46.23	53.78
EventMemAgent [35]	75.84	69.72	73.28	55.62	67.33	67.93	68.29	59.60	70.95	43.55	58.03	63.16
Ours												
VAM (Ours)	85.23	<u>76.15</u>	<u>81.90</u>	82.19	74.26	<u>73.91</u>	<u>78.94</u>	62.29	66.22	45.16	<u>57.89</u>	68.41

Table 4.1: Results on OVO-Bench [19] for Real-Time Visual Perception and Backward Tracing. The reported VAM configuration uses Gemini Embedding 2 as the embedding backbone for memory indexing and retrieval, Gemini 3 Flash as the query-time MLLM, and an initial frame sampling rate of 0.5 fps; the retained memory itself is then formed adaptively rather than stored at a constant fps. The Gemini 3 Flash baseline reported here is reproduced by our own run under the OVO-Bench protocol. The Overall column reports the mean of the two mode averages. Best model results (excluding human) are shown in **bold**, and second-best model results are underlined.

appears. Tables 4.1 and 4.2 report these results separately. For VAM, the most informative components are Real-Time Visual Perception and Backward Tracing, because they reveal whether online indexing preserves a usable memory substrate as the stream evolves and whether agentic retrieval can subsequently localise earlier evidence from that substrate.

In Table 4.1, the Real-Time Visual Perception columns correspond to Optical Character Recognition (OCR), Action Recognition (ACR), Attribute Recognition (ATR), Spatial Understanding (STU), Future Prediction (FPD), and Object Recognition (OJR), while the Backward Tracing columns correspond to Episodic Memory (EPM), Action Sequence Identification (ASI), and Hallucination Detection (HLD). In Table 4.2, the Forward Ac-

Model	Forward Active Responding			Cross-Mode Summary			Overall
	REC	SSR	CRR	RT Avg.	BT Avg.	Fwd Avg.	
Human							
Human Agents	95.48	89.67	93.56	93.20	92.33	92.90	92.81
Proprietary MLLMs							
Gemini 3 Flash [59]	35.53	74.24	61.67	72.44	62.47	57.15	64.02
GPT-4o [80]	27.58	73.21	59.40	64.46	60.75	53.40	59.54
Open-Source Offline MLLMs							
Qwen2-VL-72B [81]	38.83	64.07	45.00	61.92	56.96	49.30	56.06
LLaVA-Video-7B [82]	34.10	69.95	60.42	63.36	40.40	54.82	52.86
LLaVA-OneVision-7B [83]	25.64	67.09	58.75	64.02	43.71	50.49	52.74
Qwen2-VL-7B [81]	31.66	65.82	48.75	55.98	46.46	48.74	50.39
InternVL-V2-8B [84]	26.50	59.14	54.14	60.39	43.44	46.59	50.14
LongVU-7B [85]	12.18	69.48	60.83	57.61	35.01	47.50	46.71
Qwen3-VL-8B [42]	47.13	67.57	52.50	65.52	46.18	55.73	55.81
Open-Source Online MLLMs/Agents							
Flash-VStream-7B [86]	8.02	67.25	60.00	28.37	27.38	45.09	33.61
Dispider [88]	18.05	37.36	48.75	54.55	36.06	34.72	41.78
Streamo-7B (2fps*) [89]	30.84	57.55	82.50	67.44	49.18	56.96	57.86
ViSpeak-7B [65]	33.81	68.52	60.42	66.28	57.52	54.25	59.35
StreamForest-7B [29]	32.81	70.59	57.08	61.20	52.01	53.49	55.57
StreamAgent [66]	35.90	48.40	52.00	61.32	46.23	45.43	50.99
EventMemAgent [35]	33.67	72.02	62.08	68.29	58.03	55.92	60.75
Ours							
VAM (Ours)	22.78	58.82	65.42	78.94	57.89	49.01	61.95

Table 4.2: Results on OVO-Bench [19] for Forward Active Responding, augmented with cross-mode summaries from the main benchmark table. ‘RT Avg.’ denotes the average over Real-Time Visual Perception, ‘BT Avg.’ denotes the average over Backward Tracing, and ‘Fwd Avg.’ denotes the mean across the three Forward Active Responding tasks reported here. The final ‘Overall’ column is the mean of ‘RT Avg.’, ‘BT Avg.’, and ‘Fwd Avg.’. The reported VAM configuration uses Gemini Embedding 2 as the embedding backbone for memory indexing and retrieval, Gemini 3 Flash as the query-time MLLM, and an initial frame sampling rate of 0.5 fps; the retained memory is then formed adaptively rather than stored at a constant fps. The Gemini 3 Flash baseline reported here is reproduced by our own run under the OVO-Bench protocol. Best model results (excluding human) are shown in **bold**, and second-best model results are underlined.

tive Responding columns correspond to Repetition Event Count (REC), Sequential Steps Recognition (SSR), and Clues Reveal Responding (CRR). The additional ‘RT Avg.’, ‘BT Avg.’, and ‘Fwd Avg.’ columns expose the corresponding mode-level averages, and the final ‘Overall’ column reports their mean. Rows without reported Forward Active Responding scores are omitted from Table 4.2; this includes *SimpleStream*, for which no official forward scores are currently available. Several prior systems report OVO-Bench only partially, so the forward comparison is narrower than the main online table.

VAM’s lead on the main table reflects a more balanced profile across immediate perception and retrospective recovery than most competing methods. *SimpleStream* achieves

the strongest Real-Time average, but its Backward Tracing average drops to 54.03, whereas *Gemini 3 Flash* reaches the strongest Backward Tracing average but remains lower on the Real-Time side. VAM records a Real-Time average of 78.94 and a Backward Tracing average of 57.89, yielding the highest main-table score of 68.41. At the sub-task level, VAM is particularly notable on Spatial Understanding and Episodic Memory, which suggests that the retained memory remains useful both for understanding the current scene and for recovering earlier evidence when later questions require localisation.

4.3 MM-Lifelong Results

Long-horizon evaluation is conducted on *MM-Lifelong train@month*, which tests persistent memory over the longest temporal horizon considered in this work [3]. This setting assesses whether VAM remains searchable, temporally discriminative, and visually inspectable when relevant evidence is sparse, temporally distant, and difficult to disambiguate. In this month-scale setting, VAM attains 17.11% answer accuracy, remaining close to the strongest reported result despite the substantially heavier retrieval burden than in OVO-Bench.

This experiment is also the most expensive component of the evaluation. Relative to OVO-Bench, the month-scale setting requires substantially larger retrieval scopes, more candidate inspection, and higher end-to-end inference cost. *MM-Lifelong train@month* is particularly demanding because relevant evidence may be separated by very large temporal gaps, placing much stronger pressure on long-range retrieval and evidence verification for each query. The decision to report this month-scale setting is therefore a deliberate cost-aware evaluation choice rather than an omission of the long-horizon question.

The benchmark protocol itself is also important for interpreting the comparison correctly. In the original *MM-Lifelong* evaluation, end-to-end MLLMs are not treated as models that can absorb the full month-long stream directly; instead, they receive uniformly sparse samples up to their maximum context capacity, and the benchmark reports results under the best sparse-sampling configuration and the maximum supported context length [3]. For agentic methods, the benchmark follows official default settings, while for DeepVideoDiscovery the controller and visual model are aligned with ReMA to improve fairness [3]. Grounding is also extracted differently across paradigms: end-to-end

MLLMs are prompted to output temporal evidence intervals directly, whereas agentic systems are scored using the clue intervals surfaced during their reasoning process [3]. This means that the month-scale comparison is not between systems that all consume the same dense raw stream in one pass; it is a comparison between bounded-context sparse-sampling baselines and architectures that explicitly manage memory and retrieval over the full temporal horizon.

Table 4.3 summarises the reported *MM-Lifelong train@month* results used here to contextualise the difficulty of month-scale retrieval. Unlike long-video benchmarks that are defined mainly by clip length, MM-Lifelong is organised around persistent memory horizons at the *day*, *week*, and *month* levels, and explicitly distinguishes raw observational duration from the broader physical temporal span over which relevant evidence may be distributed [3]. The *train@month* setting is therefore intended to stress retrieval and reasoning when relevant observations may be separated by very large temporal gaps and surrounded by substantial distractor content. Here, *Ref@300* denotes quantised temporal grounding at a 300-second resolution: the benchmark discretises the timeline into 5-minute bins and scores the overlap between the predicted clue interval and the annotated clue interval at that temporal granularity [3]. The large gap between current models and the human reference illustrates how demanding this setting remains even when substantially more context is made available to the model. The corresponding frame-count comparison is reported separately in Table 4.4 below.

The table highlights two points. First, VAM attains the second-highest reported answer accuracy, only 0.51 percentage points below the strongest reported result, and substantially above all listed end-to-end MLLM baselines. Second, its *Ref@300* remains more modest than its answer accuracy. This metric design helps explain the gap: VAM is optimised to preserve sparse but inspectable visual evidence, which can be sufficient for answering a question correctly, but *Ref@300* rewards overlap with the benchmark’s annotated 300-second clue intervals rather than evidence discovery alone. As a result, retrieved images may support the right answer while still yielding limited interval-level overlap if the recovered evidence is temporally sparse, fragmented, or offset from the annotated clue span.

Method	Train@Month Acc.	Train@Month Ref@300
Human		
Human	82.5	31.2
End-to-End MLLMs		
GPT-5	10.15	1.39
Qwen3-VL-235B-A22B	9.09	0.39
Qwen3-VL-30B-A3B	8.33	0.48
Video-XL-2-8B	6.02	0.00
Video-XL-2-8B	4.89	0.09
Eagle-2.5-8B	3.76	1.59
Eagle-2.5-8B	2.07	0.71
Nemotron-v2-12B	7.52	0.19
Nemotron-v2-12B	7.71	0.18
Agentic Methods		
VideoMind-7B	5.26	1.00
LongVT-7B	5.83	1.71
DeepVideoDiscovery	4.36	2.03
ReMA with GPT-5	17.62	9.91
ReMA with Qwen3VL-A22B	14.23	<u>6.01</u>
Ours		
VAM (Ours)	<u>17.11</u>	3.65

Table 4.3: Results on MM-Lifelong train@month [3]. Train@Month Acc. denotes answer accuracy, and Train@Month Ref@300 denotes quantized temporal grounding at a 300-second resolution. Repeated model-family names are reproduced as reported in the cited benchmark results rather than being assigned unsupported variant labels here. Best model results (excluding human) are shown in **bold**, and second-best model results are underlined.

4.4 Storage Cost Analysis

The month-scale MM-Lifelong setting also makes it possible to quantify how aggressively VAM reduces the visual stream during memory construction. For the evaluated month-scale run, the original video corpus contains 11,406,923 frames in total. A fixed-rate reduction to 1 fps would still leave 380,304 frames, and even a 0.5 fps reduction would leave 190,152 frames. After online indexing and filtering, however, VAM keeps 6,876 images.

These ratios are important because they show that VAM is not just a lightly downsampled version of the input stream. In the evaluated month-scale setting, VAM keeps only 0.06% of the raw video, whereas a simple 0.5 fps policy would still keep 1.67% and a 1 fps policy would keep 3.33%. Put differently, even these coarse fixed-rate baselines would still keep $27.7\times$ to $55.3\times$ more frames than the final VAM output, while the raw stream itself is

Setting	# Stored Frames	Share of Raw Video	Multiple of VAM
Raw video frames	11,406,923	100.00%	1658.9 $\times$ more
Uniform 1 fps sampling	380,304	3.33%	55.3 $\times$ more
Uniform 0.5 fps sampling	190,152	1.67%	27.7 $\times$ more
VAM output	6,876	0.06%	1.0 $\times$

Table 4.4: Frames kept under different settings for the evaluated MM-Lifelong month-scale run. The main point is that VAM reduces the stored visual stream far more aggressively than either fixed-rate sampling or the raw input itself: the final VAM output contains only 6,876 images, compared with 190,152 frames at 0.5 fps, 380,304 frames at 1 fps, and 11,406,923 raw video frames. ‘Share of Raw Video’ reports the proportion of the original 11,406,923 video frames kept by each setting, and ‘Multiple of VAM’ reports how many times as many frames each setting keeps as the final VAM output.

1658.9 $\times$ larger. For reference, this 1 fps regime is also the relevant comparison point for frame-based systems such as EGAgent [31]: mapped onto the same month-scale stream, it would correspond to approximately 380,304 frames. In other words, the main storage benefit does not come only from selecting a lower nominal sampling rate; it comes from the subsequent boundary-sensitive filtering, which removes large amounts of redundant visual content while still preserving a searchable substrate for later evidence recovery.

This also helps explain the practical cost profile of the system. Long-horizon retrieval remains expensive because the surviving entries still have to be embedded, indexed, searched, and sometimes inspected by the MLLM at query time. At the same time, the month-scale experiment would be substantially less tractable if all 0.5 fps candidates, let alone all raw frames, had to be kept as first-class memory entries. The observed reduction from 190,152 uniformly sampled frames to 6,876 VAM images therefore provides concrete evidence that the filtering stage is doing substantive work beyond coarse frame-rate reduction alone.

4.5 Key Findings

The resulting evaluation supports two main conclusions. First, on OVO-Bench, VAM attains the highest score on the main table for Real-Time Visual Perception and Backward Tracing, with 68.41 versus 67.70 for *SimpleStream*, 67.46 for the reproduced *Gemini 3 Flash* baseline, 63.16 for *EventMemAgent*, and 61.90 for *ViSpeak*. On the broader OVO-Bench summary that also includes Forward Active Responding, VAM reaches an overall score of 61.95, establishing a new state of the art among the compared open-source agentic

systems and frameworks; this remains below *Gemini 3 Flash* at 64.02, but that baseline does not accept ultra-long video and is evaluated under bounded-context conditions. Second, on month-scale MM-Lifelong train@month, VAM attains 17.11% answer accuracy, only 0.51 percentage points below the strongest reported baseline established by *ReMA with GPT-5* at 17.62%, while remaining above *ReMA with Qwen3VL-A22B* at 14.23% and substantially above all listed end-to-end MLLM baselines, whose best reported accuracy is 10.15%. Taken together, these findings indicate that explicit, inspectable visual memory is not only compatible with online operation, but can also support strong performance across both streaming and month-scale evaluation settings.

To complement these findings, Figure 4.1 provides two larger radar-chart views over representative OVO-Bench baselines. In addition to stronger reference systems such as *SimpleStream*, *EventMemAgent*, and *ViSpeak*, the visual comparison also includes newer but weaker methods such as *ThinkStream*, *Flash-VStream*, and *Dispider*, so that the geometric spread reflects not only top-performing systems but also more recent online designs whose performance remains uneven across sub-tasks.

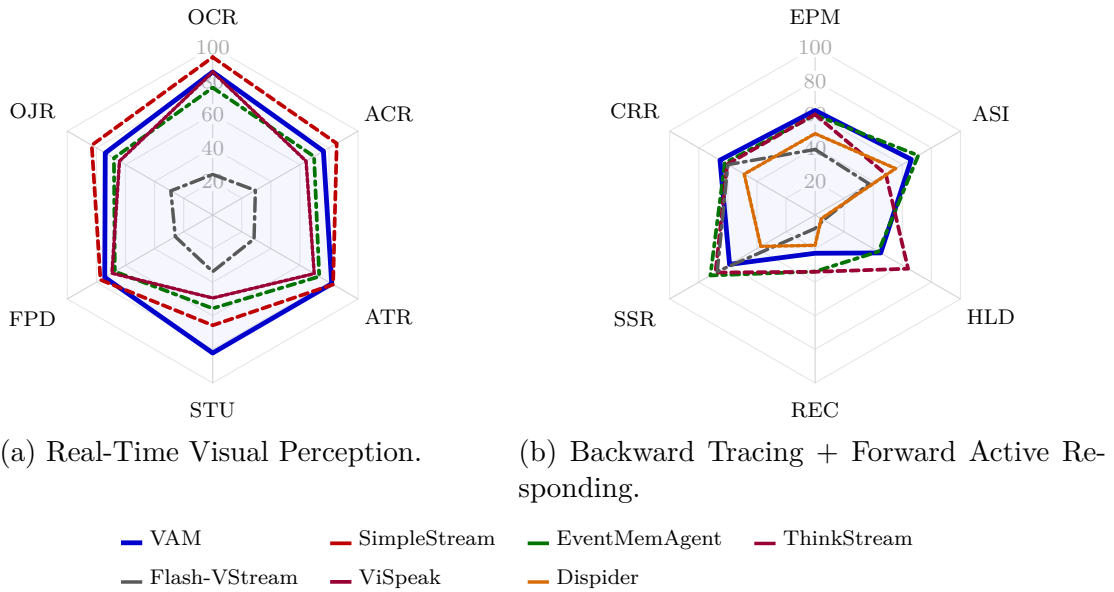


Figure 4.1: Radar-chart view of representative OVO-Bench results. The top panel focuses on Real-Time Visual Perception and includes both stronger baselines (*SimpleStream*, *EventMemAgent*) and newer but weaker systems such as *ThinkStream* and *Flash-VStream*. The bottom panel combines Backward Tracing and Forward Active Responding for methods that report both sets of scores, again mixing stronger references with newer lower-performing systems such as *Flash-VStream* and *Dispider*. This selection is intended to make the performance spread easier to read visually rather than to reproduce the full benchmark table.

In the Real-Time panel, VAM forms a broad polygon rather than a narrow spike on only one or two axes: although *SimpleStream* remains stronger on several perception tasks, VAM expands markedly on Spatial Understanding and stays competitive across Action Recognition, Attribute Recognition, and Object Recognition. In the lower panel, VAM again shows a relatively even footprint across Episodic Memory, Action Sequence Identification, Sequential Steps Recognition, and Clues Reveal Responding, whereas competing methods are generally more uneven across the six axes. *EventMemAgent*, for example, peaks more strongly on Action Sequence Identification and Sequential Steps Recognition, while *ViSpeak* is stronger on Hallucination Detection.

The Forward Active Responding setting should be interpreted differently from the other two OVO-Bench modes. Whereas Real-Time Visual Perception asks whether the model understands the current stream state and Backward Tracing asks whether it can recover earlier evidence, Forward Active Responding asks whether the system can withhold commitment until enough future evidence has actually appeared. In this sense, it is not only a test of recognition or retrieval quality, but also a test of calibration under partial evidence. VAM performs less strongly in this setting because its controller is explicitly evidence-seeking and citation-oriented: once the retrieval process surfaces observations that appear sufficient to support an answer, the agent has a tendency to treat those observations as adequate grounds for commitment, even when the stream has not yet revealed the decisive later evidence. The same evidence-first design that is advantageous for retrospective grounding can therefore become over-eager in forward tasks, where apparently supportive cues may still be incomplete. This behaviour is consistent with the gap between VAM’s strong main-table overall result and its lower Forward Active Responding average.

Taken together, the results are consistent with the central claim of this dissertation. They suggest that multimodal systems benefit when visual memory remains explicit, retrievable, and open to inspection rather than being absorbed into compressed intermediate state alone. In this respect, VAM provides a concrete and scalable approach to bridging the gap between passive video recording and interactive, intelligent analysis.

5 Conclusion

This chapter concludes the dissertation by summarising the main technical findings, discussing their broader implications, and outlining the principal limitations and future directions of the work. It returns to the central question of how online long video understanding can remain both scalable and evidentially grounded when visual streams must be indexed continuously and retrieved over extended temporal horizons. Within this setting, the proposed Visual Agentic Memory (VAM) framework has been developed as an evidence-preserving architecture that combines online indexing, hierarchical memory, and agentic retrieval. The following sections discuss what these design choices contribute, where the current system remains limited, and which directions appear most important for further development.

5.1 Discussion

The experimental results suggest three main conclusions about the architectural requirements of online long video understanding. First, the task is not reducible to retrieval alone; it depends on the coupling between online indexing and query-time search. When indexing retains excessive redundancy, memory quality degrades over time and later retrieval becomes less discriminative. Conversely, even a well-organised memory remains ineffective if query-time search cannot localise relevant evidence efficiently. Scaling to long horizons therefore requires both boundary-aware online indexing and structured search over memory rather than either component in isolation.

Second, the results show that online operation and long-horizon queryability need not be treated as mutually exclusive design goals. VAM demonstrates that a system can update memory continuously under streaming constraints while still supporting later retrieval over much longer timelines. This is reflected both in its leading result on the main OVO-Bench setting and in its competitive performance on MM-Lifelong *train@month*. This is significant because much of the prior literature implicitly resolves the design tension by prioritising one side of the problem over the other, either favouring immediate online

responsiveness or favouring longer-range reasoning after heavier consolidation.

Third, the findings underscore the importance of recoverable raw evidence within the reasoning loop. Unlike compression-heavy paradigms that retain only summaries or other abstracted memory states, VAM preserves the original visual pixels as a high-fidelity spatial substrate. This makes later visual inspection possible when fine-grained object state, spatial detail, or photometric evidence becomes critical. More broadly, the coordinated use of temporal summaries for coarse localisation and spatial evidence for frame-level verification proves essential for resolving the ambiguity of long-horizon queries while keeping the reasoning process auditable.

5.2 Practical Applications

While the core VAM framework is designed for generalised video understanding, its always-on, training-free nature naturally extends to specialised real-world applications. By treating continuous visual streams as structured, queryable memory, VAM serves as a memory infrastructure for downstream applications that require persistent environmental awareness.

5.2.1 Egocentric Vision and Wearable AR Devices

In the context of Augmented Reality (AR) and wearable devices (e.g., smart glasses), systems must process first-person (egocentric) video continuously [31]. Unlike fixed surveillance cameras, egocentric streams are characterised by rapid motion, frequent occlusions, and extreme temporal sparsity. VAM is suited for this domain; its adaptive deduplication pipeline addresses the battery and storage constraints of wearable devices by pruning quasi-static routines while preserving event boundaries. Furthermore, the agentic retrieval loop allows the AR assistant to answer long-range episodic queries by tracing back through the hierarchical memory stack without requiring constant cloud streaming of raw video.

5.2.2 3D Scene Reconstruction and Embodied Navigation

For embodied agents and robotics, 2D video understanding is often a precursor to 3D spatial awareness. VAM can act as an adaptive visual filtering stage for 3D reconstruction pipelines such as Neural Radiance Fields [90] or 3D Gaussian Splatting [91]. Unlike

many video memory frameworks that aggressively discard raw pixels, VAM explicitly retains the original RGB frames within its spatial representation. This architectural choice is critical for downstream spatial modelling, as these techniques rely on precise photometric and geometric information that cannot be recovered from a compressed temporal representation.

As illustrated in Figure 5.1, VAM concurrently maintains raw RGB frames and frame embeddings for spatial localisation, together with temporally ordered summaries and timestamps for timeline reasoning. This architecture allows arbitrary downstream applications, ranging from 3DGS / NeRF to embodied AI motor control, to query and retrieve the exact form of data they require, bridging the gap between passive continuous video logging and active application integration.

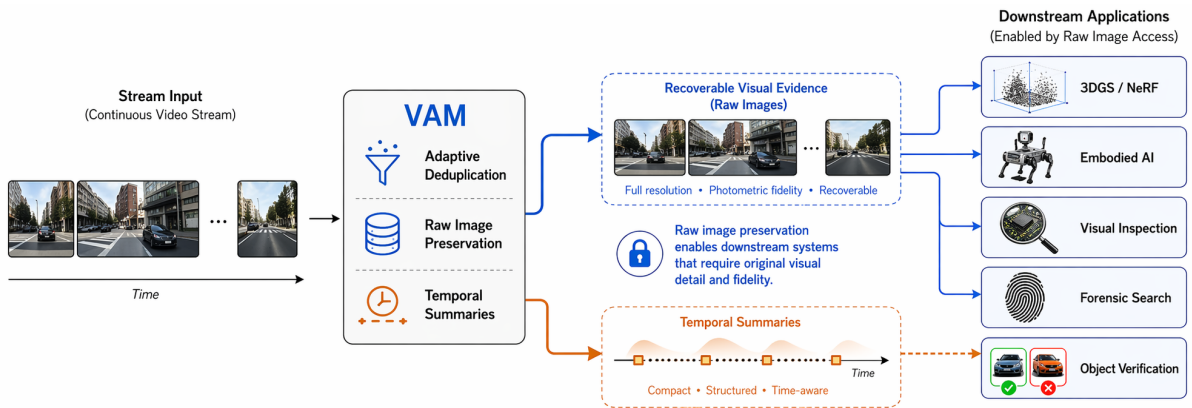


Figure 5.1: Downstream applications enabled by recoverable raw visual evidence in VAM. By preserving raw images alongside temporal summaries, the framework supports tasks that require original visual detail, including 3D reconstruction, embodied AI, visual inspection, forensic search, and object verification.

5.2.3 Continuous Security and Forensic Analysis

In long-horizon security monitoring, VAM transforms the paradigm by converting raw surveillance streams into searchable hierarchical memory with progressively compressed long-range representations. When a security event occurs, operators can use the retrieval agent in VAM to perform compositional forensic queries. The agentic retrieval loop uses both temporal retrieval and spatial retrieval to filter false positives, ensuring that forensic analyses are both highly accurate and fully auditable over extended surveillance timelines.

5.3 Limitations

While the framework demonstrates robustness across both online and longer-horizon settings, several important limitations remain. The first is a modality asymmetry in evidence preservation. VAM is designed primarily as a visual memory system: it explicitly retains uncompressed visual frames for later inspection, but it does not yet preserve original audio streams with the same level of granularity, nor does it maintain an equally rich retrieval substrate for speech, ambient sound, or other non-visual cues. This matters because many real-world queries depend on evidence that is not purely visual, such as speaker identity, verbal commitments, alarms, or environmental audio context. In such cases, the current framework may recover the relevant visual interval while still lacking the full multimodal evidence required for complete verification.

A second limitation concerns the long-term scaling behaviour of the memory itself. Although hierarchical organisation improves tractability relative to flat prompt accumulation, retrieval does not become free simply because evidence is structured. As the archive grows from hours to months and potentially to year-long deployment, the search space becomes increasingly populated with semantically similar moments, repeated routines, and visually overlapping events. Under such conditions, coarse temporal localisation may remain feasible while fine-grained disambiguation becomes progressively harder. This introduces a structural tension between coverage and selectivity: storing more evidence improves recoverability, but it also increases the burden on ranking, filtering, and verification at query time. Relatedly, the quality of later answers remains dependent on the quality of earlier indexing decisions. If an event boundary is missed, if a visually subtle state change is not retained, or if redundant content dominates the searchable memory, the downstream retrieval agent can only reason over an already imperfect substrate.

A third limitation is operational cost. The present framework relies on a multi-stage pipeline that includes continuous online indexing, embedding generation, hierarchical storage, and an iterative retrieval-and-inspection loop at query time. This design improves auditability and long-range recoverability, but it also introduces substantial inference latency and serving cost compared with simpler end-to-end or heavily compressed alternatives. The problem becomes more acute in long-horizon settings, where each query may require broad temporal search followed by repeated visual inspection over

multiple candidates. Under strict resource budgets, this makes large-scale deployment substantially more difficult than in simpler recent-context or compressed-memory systems. The evaluation scope of this dissertation also reflects that constraint, because extended-horizon experiments are considerably more expensive to run and verify than standard online benchmarks. Even so, while the reported results establish the feasibility of the architecture, they should not yet be read as evidence that the current implementation fully resolves the engineering requirements of persistent, low-cost, year-scale multimodal memory for deployment.

5.4 Ethics and Privacy

As continuous video understanding scales toward pervasive lifelogging and persistent environmental sensing, privacy and ethical deployment become first-order design constraints rather than secondary implementation concerns. The always-on nature of a system such as VAM creates substantial privacy risk because long, unbroken video streams inevitably capture sensitive personal information, bystanders who have not explicitly consented to recording, private spaces, and temporally extended behavioural patterns that may reveal health status, daily routines, relationships, or other highly sensitive attributes.

This raises several distinct ethical challenges. The first is consent. In many realistic deployment environments, the primary user of a wearable or ambient sensing system may agree to continuous recording, while nearby individuals do not. A memory system that supports long-range search could therefore amplify the privacy impact of incidental capture by making previously forgotten or hard-to-locate moments directly retrievable. The second challenge is retention and access control. Even when long-horizon memory is technically useful, unrestricted storage of raw visual evidence increases the risk of secondary misuse, unauthorised access, surveillance abuse, or retrospective behavioural profiling. The third challenge concerns representational error: an indexed memory system may appear authoritative even when its summaries, boundaries, or retrieved candidates remain incomplete, ambiguous, or contextually misleading. If such outputs are over-trusted in downstream domains such as workplace monitoring, personal assistance, or forensic review, the system could produce harms not only through privacy invasion but also through misinterpretation.

These concerns suggest that future deployments of online long video understanding should incorporate privacy-preserving indexing, explicit retention controls, user-configurable deletion policies, and consent-aware access restrictions as core architectural elements rather than optional safeguards.

5.5 Future Work

Building upon the VAM framework, several future directions appear especially important for advancing online long video understanding beyond the current feasibility stage. The first is multimodal memory expansion. The present system is intentionally centred on recoverable visual evidence, but many practically meaningful queries depend on audio, speech, alarms, environmental sound, or cross-modal temporal alignment. Extending the hierarchical memory substrate to preserve synchronised audio-visual evidence would make the framework substantially more useful in realistic settings while also reducing the current modality asymmetry identified in the limitations.

The second direction concerns efficiency and scaling. Although the current architecture demonstrates that online updating and long-horizon retrieval can coexist, this coexistence is still computationally expensive. A major next step is therefore to reduce the query-time cost of agentic retrieval without sacrificing recoverability. One promising avenue is to push more selective structuring into the indexing stage, so that downstream retrieval begins from a better organised substrate and requires fewer expensive inspection steps. Relatedly, smaller specialist models or distilled policies could take over portions of retrieval planning, candidate filtering, or routine inspection, reserving larger multimodal models for the most ambiguous cases. Such a shift would preserve the basic evidence-first philosophy of VAM while improving practicality for longer deployments and broader evaluation coverage.

The third direction is stronger memory abstraction without losing evidential grounding. At present, VAM is deliberately retrieval-first: it stores events, summaries, and frame-level evidence, then relies on the agent to recover and inspect what is needed at query time. Future systems may benefit from introducing a more explicit intermediate notion of world state, situation schema, or persistent entity memory that captures longer-range regularities across events. The challenge is to develop such abstractions without collaps-

ing the system back into an opaque compressed state that hides the original evidence. In this sense, the most promising progression is not to replace retrieval with abstraction, but to layer higher-level state modelling on top of an evidence-preserving substrate, so that causal reasoning, state tracking, and anticipatory assistance remain tethered to inspectable memory.

Finally, future work should expand both the evaluation setting and the human-facing interaction model. Current benchmarks remain limited in their coverage of month-scale deployment, interactive correction, user trust calibration, and real-world assistant behaviour under persistent use. A more complete assessment would require longitudinal evaluation, richer multimodal benchmarks, and user studies examining how people query, verify, and contest the outputs of long-horizon memory agents. From a systems perspective, this would also create opportunities to study adaptive retention policies, user-guided forgetting, and personalised retrieval control.

Overall, this work demonstrates that an online long video understanding architecture can serve as a practical foundation for evidence-grounded long-horizon visual understanding. By preserving uncompressed spatial detail alongside searchable temporal structure, VAM shows that a single evidence-preserving design can support both online responsiveness and long-horizon recoverability within one architecture. This provides a practical basis for online and long-horizon video understanding.

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A Implementation Defaults and Control Interfaces

This appendix documents the implementation-level choices that are important for faithful reproduction of the system but unnecessarily detailed for the main methodological narrative. The emphasis is therefore on concrete defaults, controller constraints, and runtime behaviour in the present software artefact.

A.1 Runtime Surfaces and Persistence

The current repository is implemented as a library-first package with optional remote access. The source code is publicly available as a GitHub repository. This repository snapshot is equivalent to the attached ZIP archive submitted with the dissertation. The implementation details documented in this appendix correspond to the following repository snapshot:

482817c6306cd125739047b3934001036a0a9629

Its principal runtime surfaces are:

- **vam-tui** provides the primary interactive interface.
- **vam-server** exposes the optional FastAPI and WebSocket surface.
- Indexed memory is persisted in SQLite by default.
- Background indexing jobs are managed in-memory during runtime, with task references retained explicitly to avoid detached asynchronous tasks.

On the optional remote interface, the principal entry points are:

- asynchronous indexing via `/api/frames/index_video_async`
- asynchronous indexing by local path via `/api/frames/index_video_async_by_path`
- interactive retrieval via WebSocket endpoint `/ws/agent`

The WebSocket session maintains lightweight keep-alive pings and supports chat reset. It does not currently support image input or runtime configuration mutation through that channel.

A.2 Reproduction Prerequisites

The software artefact is packaged as a Python project with a declared minimum requirement of Python 3.10, although in practice the repository is most straightforward to reproduce under a modern Python 3.11 environment. Because online indexing is implemented through FFmpeg-based frame extraction, FFmpeg must be installed on the host system before indexing can be executed. For full retrieval operation, a valid `OPENROUTER_API_KEY` must also be provided through a local environment file derived from `.env.example`. The default persistent store path is `data/frame_store.sqlite3`. Benchmark-specific model instantiations are reported in the evaluation section where they materially affect the experimental results, rather than being repeated here as fixed global defaults.

For persistence safety, SQLite writes are protected by an explicit file-lock layer. Database connections are opened with thread checking disabled (`check_same_thread=False`) to accommodate the asynchronous runtime, and background indexing tasks are retained by explicit task references so they are not detached during execution. At minimum, reproduction of the software artefact follows four steps. First, install the package from the repository root, either through a standard local installation or through an ephemeral `uv` execution path. Second, create a local environment file, provide `OPENROUTER_API_KEY`, and retain the default store path unless an alternative persistence target is needed. Third, launch either `vam-tui` for the primary local workflow or `vam-server` for the optional API surface. Fourth, index a local video through the TUI or through `/api/frames/index_video_async` or `/api/frames/index_video_async_by_path`, and then issue retrieval or summary requests over the resulting store. This sequence is sufficient to reproduce the main runtime behaviour discussed in the dissertation: online indexing, persistent storage, later evidence retrieval, and on-demand summary generation.

A.3 Online Indexing Parameters

The current implementation performs online indexing through an FFmpeg-based extraction stage, followed by lightweight heuristic filtering and embedding-level memory updates. In operational terms, the pipeline first reduces obviously uninformative frames using blur and low-change gates, and only then applies semantic comparison in the embedding space. Table A.1 summarises the configuration defaults that initialise this process.

Table A.1: Default online indexing parameters in the current implementation.

Parameter	Default	Role in the pipeline
Base sampling rate	0.5 FPS	Initial extraction rate before filtering.
Maximum extracted frames	None	No global hard cap by default; extraction is governed by the filtering stages.
τ_{blur}	20.0	Minimum Laplacian variance for sharpness filtering.
τ_{diff}	20.0	Mean absolute grayscale difference threshold for static-frame rejection.
τ_{ssim}	0.92	Structural redundancy threshold. Frames above this similarity are treated as near-duplicates.
τ_{hist}	0.0	Histogram-based redundancy gate; kept permissive by default in the current release.
Fallback τ_{sim}	0.88	Fallback embedding-space similarity threshold used when adaptive semantic deduplication cannot yet be estimated reliably.
Fallback τ_{event}	0.80	Fallback event-similarity threshold used when adaptive event-boundary estimation lacks sufficient history.
Maximum event duration	300.0 s	Limits how long a single event can grow before forced segmentation.

Two implementation details are especially important for interpreting Table A.1. First, the image-level gates are deliberately ordered as a cheap front-end cascade, so that blur and low-change rejection reduce load before multimodal embedding is invoked. Second, the semantic deduplication and event segmentation stages are not governed solely by fixed thresholds. In the current codebase, both are implemented as adaptive procedures over recent embedding-distance histories; the configured values shown above act as fallback thresholds when insufficient statistics are available, or when adaptive estimation fails. This distinction matters because the practical system is designed to remain online and boundary-sensitive rather than to operate as a purely static thresholding pipeline.

A.4 Implementation Notes

The current repository exposes concrete implementation defaults for reproducibility, including `google/gemini-3-flash-preview` as the default MLLM, `google/gemini-embedding-2-preview` as the default embedding model, a base sampling rate of 0.5 FPS, and the persistent store path `data/frame_store.sqlite3`. All experiments reported in this dissertation use an initial sampling rate of 0.5 FPS. These should be read as repository defaults rather than as fixed architectural commitments; benchmark-specific reporting remains the authoritative source when an experimental configuration is materially relevant.

A.5 Hierarchical Memory Defaults

The current codebase implements a three-tier hierarchical memory policy that trades spatial fidelity against temporal coverage as observations age. More recent frames are retained densely and without additional recompression, whereas older frames are thinned more aggressively and stored under stronger image compression. The default retention schedule is summarised in Table A.2.

Table A.2: Default hierarchical memory configuration.

Tier	Time Window	Min Gap (s)	Max Side (px)	JPEG Quality
recent	$t \leq 1$ hour	1.0	—	No additional recompression
mid	$1 \text{ hour} < t \leq 24$ hours	20.0	768	70
long	$t > 24$ hours	120.0	512	45

The retention policy contains several implementation-level safeguards that are relevant to reproducibility. The earliest and latest available frames are always preserved, and event-linked frames in the long-term tier are retained explicitly rather than being subjected to ordinary bucket-based thinning. Compression is applied only after the keep set has been determined, so the retention schedule and the image-compression schedule remain logically separate. This design preserves a minimal set of temporally anchored evidence while still keeping long-horizon storage growth manageable.

A.6 Agentic Retrieval Constraints

The retrieval controller is implemented with strict schema validation rather than free-form prompting alone. Concretely, action payloads are parsed through bounded Pydantic schemas with extra keys forbidden, ensuring that the MLLM operates within an explicit control interface. At implementation level, these schemas inherit from a strict base model with extra keys forbidden globally, so controller outputs are validated against an executable interface rather than loosely interpreted after generation. Table A.3 lists the principal action- and field-level constraints exposed by the current controller.

Table A.3: Key constraints in the current agentic retrieval schema.

Action / Field	Constraint	Meaning
<code>search.top_k</code>	1 to 200	Maximum number of candidates retrieved per query.
<code>search.inspect_k</code>	0 to 50	Number of top candidates passed to Visual Inspection; clipped to <code>top_k</code> .
<code>search.threshold</code>	≥ 0.0	Minimum retrieval threshold; default schema value is 0.65.
<code>inspect.max_frames</code>	1 to 16	Maximum number of frames examined in a direct Visual Inspection step.
<code>summarize.granularity_seconds</code>	(0, 86400]	Resolution of reusable summary windows.
<code>time_range.mode</code>	{relative, absolute, auto}	Supports both uploaded videos and absolute live/history timestamps.
<code>anchor.before_seconds</code>	[0, 604800]	Bounds event-relative windows to at most seven days on each side of the anchor.
<code>anchor.after_seconds</code>		
<code>anchor.top_k</code>	1 to 50; 0 to 20	Controls anchor-candidate retrieval and subsequent inspection budget; <code>inspect_k</code> is clipped to <code>top_k</code> .
<code>anchor.inspect_k</code>		
<code>anchor.occurrence_index</code>	1 to 100	Allows ordinal reference to repeated anchor events when such indexing is needed.
<code>anchor.verification_prompt</code>	optional, max 400 chars	Supports strict anchor confirmation without allowing arbitrarily long side instructions.
<code>sources</code>	subset of {frame, event, summary}	Allows the planner to explicitly target different memory structures.
<code>summary_filter</code>	optional	Restricts summary retrieval by structure label or granularity.

The most important architectural implication of this schema is that the controller is not restricted to a single search-inspect pattern. Instead, it can alternate among `search`,

inspect, **summarize**, and final response generation, depending on the evidence state. In addition, the time-range schema supports both direct intervals and anchored temporal reasoning, including windows defined relative to a queried event or intervals delimited by two previously returned references. These constraints give the planner a richer control surface while keeping the executable action space explicit and auditable.

B Prompt and Control Logic

This section summarises the prompt-level control logic implemented in the current code-base. The purpose is not to analyse prompt phrasing as a contribution in itself, but to make explicit the control contracts that mediate routing, retrieval, inspection, and summary writing.

B.1 Top-Level Router Prompt

At the WebSocket layer, the system first invokes a lightweight router whose sole responsibility is to choose exactly one next step for the incoming user utterance. The admissible outputs are:

```
1 1) {"type":"final","text":"..."}
2 2) {"type":"tool","name":"retrieve","args":{"question":"..."}}
3 3) {"type":"tool","name":"summarize","args":{"
4     "min_time":0.0,
5     "max_time":1800.0,
6     "time_mode":"relative",
7     "granularity_seconds":60.0,
8     "summary_structure":"optional_tag",
9     "prompt":"..."
10  }}
```

This router enforces an architectural distinction between ordinary factual querying and explicit summary creation. A normal information-seeking question is routed to **retrieve**, whereas **summarize** is reserved for constructing persistent summary documents over a specified time span. The summary branch therefore functions as a memory-writing operation rather than as a mere alternative answer format. At implementation level, this top-level router exposes only two tool calls, **retrieve** and **summarize**, in addition to direct final responses. These outputs are validated through strict schemas with extra keys forbidden, so the router cannot emit arbitrary tool names or loosely structured arguments.

In particular, `retrieve` accepts exactly one non-empty field, `question`, while `summarize` accepts a bounded time-range specification together with positive `granularity_seconds`, a non-empty prompt, and an optional normalised `summary_structure` tag.

B.2 Planner Prompt for Agentic Retrieval

Once retrieval has been selected, a separate planner prompt governs the multi-turn agentic retrieval loop. In the current implementation, the planner may emit four actions:

```
1 {"action": "answer", ...}
2 {"action": "search", ...}
3 {"action": "inspect", ...}
4 {"action": "summarize", ...}
```

The key code-level characteristics of this planner are as follows:

- **Search** is the default action for ordinary question answering.
- **Inspect** is a direct Visual Inspection step used when the relevant reference or time interval is already known.
- **Summarize** creates persistent summary documents rather than merely producing an ephemeral answer.
- **Anchored time ranges** allow the planner to resolve queries such as “before X”, “after Y”, or “between A and B” using either query-based anchors or previously returned references.

Collectively, these actions define a controller that separates semantic retrieval, direct evidence inspection, and reusable summary construction into distinct operations. This separation is important because it makes explicit when the system is searching over memory, when it is visually verifying already localised evidence, and when it is writing new persistent summary artefacts back into memory. The planner interface is also more structured than the abbreviated action list above may suggest. In the current code-base, `search` includes one or more query objects with bounded `top_k`, `inspect_k`, and threshold values, and may further specify retrieval `sources`, a `summary_filter`, a prior `visual_ref`, optional joint inspection, and a dedicated `inspection_prompt`. `Inspect`

requires a non-empty prompt together with either an explicit `time_range` or a previously returned reference, and caps `max_frames` at a small fixed upper bound. `Summary Creation` (`summarize`) requires a valid `time_range`, positive `granularity_seconds`, a non-empty prompt, and an optional `summary_structure`. `Answer` may additionally cite a supporting `best_ref`, expressed through the same turn-indexed reference format used elsewhere in the controller.

The time-range interface is itself hierarchical. A planner step may either specify direct numeric bounds (`min`, `max`, and `mode`) or an anchor-based constraint. For event-relative reasoning, an anchor may include a query or query variants, a prior reference, bounded `before_seconds` and `after_seconds`, source restrictions, grouped candidate-source strategies, an occurrence index, and an optional strict `verification_prompt`. The schema forbids mixing a single anchor with paired `start_anchor/end_anchor` boundaries, and inspect budgets are clipped so they cannot exceed the corresponding retrieval budgets. These implementation details matter because they show that the controller is not merely prompted to “reason step by step”, but is constrained to operate through an executable retrieval protocol with typed references, bounded search budgets, and explicit temporal operators.

B.3 Event Document Prompt

When an event is finalised during online indexing, the system invokes a dedicated prompt to write a retrieval-oriented event document. The prompt can be summarised as follows:

```
1 You are building a retrieval document for visual memory.
2 Given a completed event, use the full event video as the primary
3 source when available.
4 Write a detailed, factual memory document.
5 Requirements:
6 - include the time range explicitly
7 - describe people, objects, actions, scene changes, and on-screen
  text
8 - prefer a few short natural lines when there are multiple
  distinct observations
9 - stay objective
```

In the present implementation, the output of this stage is best understood not as a generic transcript, but as a retrieval-oriented memory document for a temporally bounded event. When available, the MLLM is given the event video clip cropped from the source video over the event interval together with a small set of representative retained frames and temporal metadata; only when video input is unavailable does the implementation fall back to the representative frames and metadata alone. The prompt explicitly encourages factual, time-explicit, and retrieval-friendly phrasing so that the resulting document can later support temporal search and downstream reasoning.

B.4 Reusable Summary Prompt

The summarisation tool uses a separate prompt because its output is not an event document, but a reusable summary memory note over a requested time window:

```
1 You are building reusable timeline summaries for a video-memory
   system.
2 You will receive a time window, some event documents that overlap
3 that window, and representative frames.
4 Write a factual retrieval note for that window.
5 Requirements:
6 - mention the time window clearly
7 - prefer concrete observations over speculation
8 - synthesise both the provided document context and visible
   frames
9 - make the summary retrieval-friendly for later QA
```

This separation between event documents and reusable timeline summaries is central to the memory design. Event documents record localised segments produced during online indexing, whereas summary documents are generated on demand to create coarser, reusable abstractions over user-specified intervals.

A further implementation detail affects summary materialisation. In the current code-base, the effective summary bucket size is enforced as $\max(5 \text{ s}, \text{granularity_seconds})$, overlapping event documents are included by default when constructing each summary window, and at most six representative frames are passed to the summarisation call for

that window. These choices bound prompt size while preserving enough visual evidence for later retrieval.

B.5 Anchor Verification Prompt

For ambiguous event-relative queries, the system may invoke a strict anchor verifier before committing to an anchored time span. The verifier is instructed as follows:

```
1 You are validating whether a short sequence of frames contains a
2 requested anchor event.
3 Be strict. Do not mark a match unless the event itself is
   visually
4 supported.
5 Reply with ONLY one JSON object:
6 {"match": true|false, "confidence": 0.0-1.0,
7  "observed_event": "...", "reason": "..."}

```

This verification step is particularly important for queries involving repeated events, ordinal references, or event-relative temporal windows. By requiring an explicit visual confirmation before anchoring the downstream search interval, the controller reduces the risk that an imprecise anchor candidate will propagate error into subsequent retrieval turns.

B.6 WebSocket Control Surface

The WebSocket endpoint at `/ws/agent` implements a lightweight interactive control layer above the router and planner. Inbound payloads primarily use `text` messages for ordinary interaction, but the endpoint also accepts explicit `tool` invocations for `retrieve` and `summarize`, as well as `reset` for clearing chat history. Outbound messages include `info`, `ping`, `user_text`, `tool_use`, `final`, and `error`, allowing clients to distinguish status updates, keep-alive traffic, explicit tool execution, and terminal answers. This endpoint does not currently support image messages or runtime configuration mutation through the same channel. The practical significance of this design is that the interactive API is not just a thin text wrapper around the model: it exposes a bounded message protocol that mirrors the tool-constrained architecture described above.